

LEXTAX AI: A BUSINESS COMPLIANCE INTELLIGENCE ENGINE (BCIE) FOR UNIFIED LEGAL, GST AND VENDOR RISK AGGREGATION

A PROJECT REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

Information related to compliance within a small or medium enterprise is often dispersed across separate systems: legal contracts are saved as documents, GST invoices are handled via accounting software, vendor records are monitored informally or not tracked at all, and deadlines for contract renewals or filings are maintained in spreadsheets. Current automated compliance-checking methods, including recent efforts that integrate large language models with rule-based validation, typically function at the level of individual documents or checks and fail to link risk signals across these sources.

This paper presents LexTax AI, a prototype Business Compliance Intelligence Platform built around a central Business Compliance Intelligence Engine (BCIE). BCIE receives structured risk signals from six interconnected modules — legal document classification and clause analysis, GST invoice validation with duplicate and Input Tax Credit (ITC) checks, GST liability prediction, vendor risk profiling, and contract lifecycle monitoring — and combines them into a single, explainable business-level compliance score with prioritized recommended actions.

Instead of suggesting a new technique for document classification or invoice validation, this study examines if independently generated module-level risk signals can be normalized, combined, and explained systematically at the business level, and if viewing compliance as an interconnected workflow reveals risks that separate per-module checks might overlook. The prototype was evaluated across its implemented modules for functional correctness, signal-propagation consistency, and whether BCIE's stated reasons for a given risk level matched its underlying inputs.

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LIST OF ABBREVIATIONS

ABBREVIATION EXPANSION		ABBREVIATION EXPANSION	
AI	Artificial Intelligence	LLM	Large Language Model
API	Application Programming Interface	MAE	Mean Absolute Error
BCIE	Business Compliance Intelligence Engine	MAPE	Mean Absolute Percentage Error
BCS	Business Compliance Score (0 to 100)	MSME	Micro, Small, & Medium Enterprises
BERT	Bidirectional Encoder Representations	NDA	Non-Disclosure Agreement
CBIC	Central Board of Indirect Taxes	NLI	Natural Language Inference
CGST	Central Goods and Services Tax	NLP	Natural Language Processing
CLM	Contract Lifecycle Management	OCR	Optical Character Recognition
CRDT	Conflict-free Replicated Data Type	OLS	Ordinary Least Squares
CUAD	Contract Understanding Atticus Dataset	PAN	Permanent Account Number
DFD	Data Flow Diagram	PDF	Portable Document Format
DOCX	Word OpenXML Document	RAM	Random Access Memory
ERP	Enterprise Resource Planning	REST	Representational State Transfer
FAISS	Facebook AI Similarity Search	RMSE	Root Mean Square Error
GST	Goods and Services Tax	SBERT	Sentence-BERT
GSTIN	GST Identification Number	SGST	State Goods and Services Tax
GSTN	GST Network	SHAP	SHapley Additive exPlanations
HSN	Harmonized System of Nomenclature	TDS	Tax Deducted at Source
IGST	Integrated Goods and Services Tax	UI	User Interface
ITC	Input Tax Credit	UUID	Universally Unique Identifier
JSON	JavaScript Object Notation	XGBoost	eXtreme Gradient Boosting

CHAPTER 1

INTRODUCTION

1.1 GENERAL

Regulatory, financial, and legal compliance forms the operational backbone of modern enterprise governance. Within corporate entities and Micro, Small, and Medium Enterprises (MSMEs), statutory adherence is governed by a complex array of legal enactments, indirect taxation mandates, counterparty agreements, and strict statutory filing calendars. In practice, however, compliance management remains severely fragmented across disparate operational silos. Commercial agreements such as Non-Disclosure Agreements (NDAs), Master Service Agreements (MSAs), and vendor contracts are archived as unstructured PDF documents in decentralized storage. Simultaneously, Goods and Services Tax (GST) invoices are processed independently through standalone accounting software such as Tally or custom ERP tools. Counterparty vendor reliability is evaluated informally without systematic error logging, and crucial statutory deadlines for GST return submissions are monitored through uncoordinated desktop spreadsheets.

This traditional siloed paradigm operates on the assumption that compliance can be effectively maintained through isolated post-facto audits. However, individual documents and financial vouchers do not exist in isolation; they represent interconnected operational nodes within an overarching business ecosystem. A contractual failure, such as an omitted indemnification covenant or an unmonitored expiration date, directly impacts commercial risk. Concurrently, an inadvertent tax arithmetic discrepancy or a failure to cross-verify supplier filings under Section 16(2)(aa) of the Central Goods and Services Tax (CGST) Act can trigger immediate Input Tax Credit (ITC) disallowances, mandatory cash reversals, and severe statutory interest penalties.

As indirect tax regulations undergo continuous digital reform—exemplified by GST Council rate rationalizations, mandatory e-invoicing thresholds, and algorithmic return cross-matching—the administrative burden on corporate finance and legal teams has multiplied. Enterprises face intense operational friction between rapid business execution and rigorous statutory compliance, exposing corporate balance sheets to avoidable litigation, recovery notices, and severe reputational damage.

LexTax AI introduces an enterprise-grade Business Compliance Intelligence Platform architected around a central Business Compliance Intelligence Engine (BCIE). A foundational design tenet of the platform is the architectural decoupling of deterministic statutory compliance rules from learned analytical machine learning models. Compliance verification in corporate taxation demands absolute numerical fidelity and auditability. Tax arithmetic, statutory slab splits (CGST, SGST, IGST), duplicate invoice detection, and statutory due dates must adhere strictly to codified legal provisions, leaving zero margin for probabilistic variance or model hallucination. Conversely, complex cognitive operations—such as classifying natural language contract types, extracting semantic clauses, detecting subtle contractual contradictions, and forecasting future monthly tax liabilities—benefit immensely from modern natural language processing (NLP), transformer embeddings, and tabular ensemble regression.

By orchestrating these capabilities into a unified pipeline, LexTax AI transforms compliance from a reactive, manual burden into a proactive, automated, and explainable governance mechanism. The platform ingests heterogeneous compliance signals from six interconnected modules, normalizes disparate findings into a bounded $[0, 1]$ numerical space, and deterministically aggregates them into an interpretable Business Compliance Score (BCS) ranging from 0 to 100.

1.2 PROBLEM STATEMENT

Existing enterprise compliance mechanisms function almost exclusively at the level of individual documents and isolated transactions. When an accounting department processes a vendor invoice, validation is typically restricted to basic arithmetic and voucher data entry. Conventional tools do not cross-reference the invoice against the governing contract to verify agreed payment terms, nor do they screen against the counterparty's historical error frequency or verify whether the vendor has timely uploaded their outward supplies in GSTR-1 to ensure Input Tax Credit eligibility.

This structural blind spot prevents organizations from detecting multi-vector risks where individually benign issues compound into enterprise-threatening statutory failures. A delayed vendor tax filing combined with an expiring master service agreement and an overcharged GST rate slab creates an institutional vulnerability that conventional isolated software packages fail to recognize.

Under recent Indian indirect tax reforms—including the rate rationalizations recommended by the 53rd and 54th GST Council meetings and strict statutory restrictions under Section 16(2)(aa) of the CGST Act—the financial liabilities of compliance oversights have escalated dramatically. For example, billing an industrial service at a legacy 28% rate instead of the rationalized 18% slab results in immediate cash overcharge, while claiming ITC on invoices not auto-populated in GSTR-2B attracts mandatory tax recovery alongside an 18% per annum statutory interest charge. In parallel, managing corporate agreements without automated clause verification exposes organizations to uncapped indemnities, contradictory termination terms, and disputes governed by unfavorable jurisdictions.

Crucially, there is an absence of an integrated mechanism for converting heterogeneous compliance outputs into a unified, explainable business-level risk representation. Without centralized risk aggregation, enterprise executives lack real-time visibility into overall corporate compliance posture, leaving organizations vulnerable to compounding operational and statutory penalties.

1.3 OBJECTIVES

The primary technical and operational objectives of the LexTax AI project are formulated as follows:

- Design and develop a unified enterprise compliance intelligence platform that breaks down compliance silos across legal agreements, GST invoices, vendor profiles, and statutory filing calendars.
- Implement an automated Legal Document Intelligence pipeline utilizing natural language processing and the 41-clause CUAD taxonomy to extract clauses, detect missing provisions, and screen for intra-contract contradictions.
- Develop a statutory-compliant AI Contract Generator capable of dynamically synthesizing enforceable legal agreements embedded with mandatory dispute resolution and bilateral indemnification guardrails.
- Build an automated Optical Character Recognition (OCR) and layout parsing pipeline to ingest physical invoices and digital PDF vouchers with high token fidelity.
- Synthesize dynamic risk mitigation strategies by mapping raw numerical compliance signals directly into actionable, prioritized corporate governance directives.

- Formulate a deterministic GST Invoice Validation engine to verify line-item tax arithmetic, intra-state (CGST/SGST) and inter-state (IGST) bifurcations, and detect stale tax rate applications post-September 2025 reforms.
- Implement dual-tier duplicate invoice screening utilizing cryptographic SHA-256 exact hashing alongside fuzzy date (± 7 days) and amount ($\pm ₹5$) matching windows to prevent fraudulent double billing.
- Train, evaluate, and benchmark tabular machine learning time-series regression models across 48 months of historical tax ledgers to forecast net monthly GST liabilities with bounded confidence intervals.
- Develop a multi-factor Vendor Risk Radar that profiles counterparties based on invoice error rates, duplicate voucher submissions, and GSTR-1 filing consistency.
- Architect the central BCIE aggregation engine to mathematically normalize disparate risk findings into a bounded $[0, 1]$ range, apply a transparent weighted-sum model, compute the Business Compliance Score (0–100), and dynamically generate prioritized, traceable remediation actions.

1.4 EXISTING SYSTEM

Traditional corporate compliance workflows rely upon disparate manual procedures and disconnected software packages. Accounting personnel record tax invoices into standalone ledger software such as Tally ERP. While these tools calculate mathematical totals based on user input, they lack automated OCR ingestion for physical invoices, cannot detect whether an applied tax rate is obsolete under recent statutory amendments, and do not cross-reference vendor tax filing compliance.

Legal document review in the existing framework is conducted manually by legal counsel. Each contract must be read end-to-end to verify that vital covenants are present and uncompromised. This manual review process is highly time-consuming, expensive, and susceptible to human error, often leading to overlooked clause contradictions. Furthermore, statutory filing deadlines are manually transcribed into static spreadsheets, leading to missed filing windows and late fee penalties.

Crucially, existing systems lack cross-functional intelligence. An accounts payable clerk entering a supplier invoice has no visibility into whether the governing master agreement is active or whether the supplier has a history of submitting duplicate vouchers. This complete absence of unified risk tracking results in recurring cash leakages, delayed audits, and severe regulatory non-compliance.

1.5 PROPOSED SYSTEM

The proposed LexTax AI platform overcomes the fundamental limitations of the existing fragmented paradigm by introducing an integrated, multi-tier compliance intelligence architecture. The platform combines deep natural language understanding, deterministic statutory verification, computer vision OCR, predictive machine learning, and centralized mathematical risk aggregation into a single cohesive enterprise solution. Figure 1.1 illustrates the structural comparison between the conventional manual process and the LexTax AI platform.

Aspect	Existing Manual Process	LexTax AI Platform
Tools required	Word (drafting) + Tally (GST) + Excel (tracking) + external lawyer review	Single unified platform
Legal document review	Manual, lawyer-dependent, slow to turn around	AI-assisted drafting and rule-based clause/missing-clause review (professional review still recommended)
GST validation	Manual calculation, error-prone	Deterministic rule engine, checked at entry time
Vendor risk visibility	Not tracked systematically	Aggregated, persistent vendor risk profile
Cross-source risk view	Fragmented across tools, no linkage between them	Unified BCIE compliance score with ranked, explained reasons
Historical trend tracking	Manual spreadsheet upkeep	Persisted, queryable history per business

Figure 1.1 Structural Comparison: Existing Manual vs. LexTax AI Platform

By establishing an architectural separation between deterministic statutory rules and learned machine learning models, LexTax AI guarantees that tax arithmetic and regulatory checks remain provably accurate and auditable, while leveraging AI for complex semantic analysis and time-series forecasting. The central BCIE aggregation layer provides enterprise executives with a unified, real-time dashboard displaying multi-vector radar charts, overall compliance posture, and prioritized remediation directives.

Key Operational Innovations: The platform delivers four breakthrough capabilities: (1) **Cross-Vector Synthesis:** linking contractual penalties directly to vendor invoicing discrepancies; (2) **Zero-Variance Statutory Execution:** immutable rule enforcement for tax slabs; (3) **Proactive Cash Flow Shielding:** forecasting upcoming monthly GST liabilities to prevent liquidity crunches; and (4) **Natural-Language Remediation:** translating raw mathematical error signals into executive legal directives.

Scalable Cloud Deployment Paradigm: The architecture supports containerized microservice deployments orchestrated via Kubernetes, ensuring horizontal auto-scaling during high-throughput tax filing weeks while maintaining zero downtime and complete statutory data isolation.

Report Structure: Chapter 2 reviews literature across legal NLP, taxation, and ML. Chapter 3 specifies system architecture, persistence, and APIs. Chapter 4 provides module details and the BCIE model. Chapter 5 details experimental evaluation and the Vendor ABC worked case. Finally, Chapter 6 presents conclusion and future work.

CHAPTER 2

LITERATURE SURVEY

2.1 LITERATURE REVIEW

2.1.1 Legal Document Analysis and LLM Compliance Checking

Sharma and Gupta [1] analyzed the application of large language models and prompt orchestration frameworks for automated compliance checking of unstructured business documents. Their work demonstrated that zero-shot and few-shot prompting techniques could identify standard contractual violations across enterprise agreements. Chalkidis et al. [2] introduced LEGAL-BERT, demonstrating that pre-training transformer architectures on specialized legal corpora (including legislation, court filings, and contracts) yields substantial performance gains over general-domain BERT in legal entity recognition and classification.

The relevance of this work to the present project is that it contributes one part of the overall pipeline—legal semantic extraction, transformer representation, deterministic verification, or human interaction. The project does not claim novelty for any individual NLP component; its contribution lies in integrating these natural language capabilities into an automated compliance pipeline and exposing normalized legal risk signals to a central business-level intelligence engine.

2.1.2 Contract Clause Extraction and the CUAD Benchmark

Hendrycks et al. [3] established the Contract Understanding Atticus Dataset (CUAD), providing a foundational benchmark of over 13,000 expertly annotated contract labels spanning 41 distinct clause types across commercial agreements. Their findings highlighted that specialized deep learning models achieve high accuracy on standard covenants but experience reduced precision on subtle wording variations and domain-specific carve-outs.

In enterprise contracts, missing or ambiguously drafted covenants represent a primary source of legal exposure. Automated extraction models must therefore exhibit high recall to ensure that critical indemnification, non-compete, and confidentiality obligations are reliably flagged for legal counsel before formal signing.

The relevance of this work to the present project is that CUAD provides the standardized clause taxonomy adopted by LexTax AI's Legal Document Intelligence Module. Rather than reinventing contractual categories, the project utilizes the 41-clause checklist to detect missing mandatory covenants (such as limitation of liability and mutual indemnification) and maps these omissions into a bounded risk penalty.

2.1.3 Semantic Contradiction and Inconsistency Detection

Wadden et al. [4] investigated factual inconsistency and contradiction detection within specialized technical texts. In legal agreements, contradictions frequently arise between general terms and specific exhibits, such as an uncapped indemnity clause conflicting with a strict liability limitation cap. Bowman et al. [5] and Williams et al. [6] developed large-scale Natural Language Inference (NLI) benchmarks (SNLI and MultiNLI) that formalize the classification of sentence pairs into entailment, neutral, and contradiction relationships.

The relevance of this work to the present project is that it motivates LexTax AI's intra-document contradiction screening. By comparing extracted clause pairs across related contractual topics, the platform identifies potential legal ambiguities and flags them for human legal review before execution, preventing costly commercial litigation.

Contractual Contradiction Topology: Commercial contracts exhibit complex compositional semantics where covenants governing liability, liquidated damages, and termination rights are distributed across disparate schedules. Cross-encoder models fine-tuned on legal inference allow LexTax AI to compute directional entailment probabilities, identifying reciprocal clause incompatibilities that simple lexical parsers cannot detect.

2.1.4 Dense Semantic Retrieval and Precedent Search

Reimers and Gurevych [7] developed Sentence-BERT (SBERT), utilizing Siamese and triplet network structures to generate semantically meaningful sentence embeddings that can be compared using cosine similarity. Johnson, Douze, and Jégou [8] introduced FAISS (Facebook AI Similarity Search), enabling billion-scale dense vector indexing and sub-millisecond similarity retrieval.

Dense vector representations allow semantic comparison that transcends superficial keyword matching, enabling corporate systems to identify conceptually identical covenants drafted using divergent legal terminology. By mapping clauses into high-dimensional vector space, the search engine indexes semantic intent rather than literal strings, drastically accelerating contract drafting.

The relevance of this work to the present project is that SBERT and FAISS provide the underlying technology for LexTax AI's Legal Precedent Search and clause recommendation engine. When an enterprise user audits or drafts a contract, the platform retrieves semantically aligned, legally vetted clauses from a curated repository in real time, accelerating document authoring.

2.1.5 Statutory Indirect Taxation and Deterministic Rule Verification

Purohit [9] and Rao [10] conducted extensive empirical analyses of India's Goods and Services Tax (GST) architecture, examining the operational complexities of multi-tier tax slabs, Input Tax Credit (ITC) tracking, and supply-chain tax cascading. Bird [11] demonstrated that indirect tax compliance cannot rely on probabilistic estimation; statutory authorities require exact numerical fidelity, auditable line-item arithmetic, and strict adherence to codified tax rules.

Statutory Mechanics of Input Tax Credit: Under Section 16(2)(aa) of the CGST Act, a recipient is legally entitled to claim input tax credit only if the supplier has furnished the invoice details in their GSTR-1 and such details are formally communicated to the recipient in Form GSTR-2B. Inadvertent claims on unreflected invoices attract immediate demand notices under Section 73/74, coupled with mandatory 18% per annum statutory interest and steep penalties, imposing severe liquidity strains on enterprise balance sheets.

The relevance of this work to the present project is that it justifies the architectural separation of deterministic validation from learned AI models. Operations such as tax arithmetic, CGST/SGST/IGST splits, and stale rate detection under recent GST Council directives are implemented through immutable programmatic rules rather than probabilistic models.

2.1.6 Financial Document Processing, OCR, and Fraud Screening

Cui et al. [12] developed Document AI frameworks integrating visual layout and textual features for automated form parsing. In parallel, financial fraud literature establishes that duplicate invoice submissions—both exact and fuzzy near-duplicates—constitute a primary source of corporate cash-drain in commercial accounting departments.

Deep Layout Analysis for Multi-Page Invoices: Real-world commercial invoices feature irregular table borders, dense numeric grids, and varying key-value spatial arrangements. Combining deep OCR with cryptographic hashing establishes an immutable defense against double-billing fraud, ensuring that item descriptions, quantities, and HSN codes are not conflated.

The relevance of this work to the present project is that it forms the document ingestion and invoice fraud detection layer. LexTax AI deploys deep OCR for physical invoice digitization and executes dual-layer screening (cryptographic SHA-256 for exact duplicates and fuzzy date/amount windows for disguised duplicates) to safeguard corporate disbursements.

2.1.7 Time-Series Regression for Tax Cash Flow Forecasting

Time-series literature establishes that accurate forecasting of tax liabilities requires capturing quarterly seasonality and macroeconomic cycles. In enterprise finance, machine learning regression models have demonstrated high predictive accuracy on tabular ledger data by mapping historical remittances to upcoming statutory obligations.

The relevance of this work to the present project is that proactive compliance demands forward-looking visibility. By training regression models across 48 months of corporate ledgers, LexTax AI forecasts net monthly GST liabilities, enabling finance teams to optimize working capital and ensure timely statutory payments.

2.1.8 Counterparty Vendor Risk and Explainable Decision Systems

Explainable AI research demonstrates that automated decisions in critical business domains must remain transparent and auditable. Vendor risk literature confirms that counterparty compliance failures—such as erroneous invoicing or delayed tax filing—directly jeopardize the buyer's tax credit eligibility under Section 16(2)(aa) of the CGST Act.

The relevance of this work to the present project is that LexTax AI synthesizes counterparty performance into an auditable Vendor Risk Radar and ensures that all composite risk scores are decomposable into human-understandable contributing factors. Tracking vendor reliability over longitudinal transaction windows prevents repeated cash flow penalties.

2.2 COMPARISON AND DISCUSSION

To rigorously position LexTax AI within the existing research and commercial landscape, Table 2.1 presents a comprehensive comparison of related compliance and document automation approaches across six functional dimensions.

Approach / System	Primary Strength	Limitation Relative to This Project
Standalone Accounting Software (e.g., Tally ERP)	Robust double-entry bookkeeping and local ledger computation	No automated contract analysis, zero vendor risk profiling, and manual deadline tracking
Domain-Specific Legal NLP (e.g., LEGAL-BERT, CUAD)	High-accuracy legal clause classification and covenant extraction	Operates strictly on isolated legal text; completely detached from GST and financial ledgers
Document AI & OCR Systems (e.g., PaddleOCR, LayoutLM)	Effective extraction of text boxes and structured key-value pairs	Restricted to text recognition; lacks statutory compliance verification and risk aggregation
Tabular ML Forecasting (e.g., XGBoost, Random Forest)	Accurate time-series prediction of corporate cash flow and revenue	Produces isolated forecasts without statutory tax rule validation or legal audit capability
Dedicated Contract Lifecycle Management (CLM)	Structured tracking of contract expiration and milestone alerts	No integration with tax filing status, invoice validation, or statutory compliance calendars
Generic LLM Prompting (e.g., GPT-4 Zero-Shot)	Broad natural language understanding and flexible summary generation	High risk of probabilistic hallucination in statutory tax arithmetic; black-box reasoning
Proposed LexTax AI Platform	Unified multi-vector compliance intelligence engine (BCIE) with explainable aggregation	Requires structured document feeds and enterprise persistence infrastructure

The comparative analysis reveals a critical structural gap in existing enterprise software: while specialized tools exist for individual tasks (bookkeeping, legal drafting, OCR, or time-series prediction), no existing platform provides a mathematically rigorous, explainable mechanism for aggregating heterogeneous compliance outputs into a unified business posture.

Cross-Domain Synthesis Failure: Disconnected point solutions force enterprise risk officers to manually reconcile disjointed reports. Without an underlying mathematical normalization framework, comparing a high-severity contract indemnification gap against a minor recurring GST tax mismatch remains fundamentally arbitrary and prone to subjective oversight.

Furthermore, existing approaches fail to make a clear architectural distinction between deterministic statutory compliance and learned analytical modeling. Attempting to solve statutory tax calculations with large language models inevitably introduces probabilistic errors and non-reproducible behavior that violates corporate auditing standards. Conversely, relying solely on rigid rule-based systems prevents organizations from interpreting natural language contracts or anticipating future tax liabilities.

LexTax AI resolves this dichotomy by establishing a decoupled architecture. Deterministic engines enforce statutory tax rates, exact arithmetic, and filing deadlines with zero variance, while learned AI components handle clause semantics, vector search, and time-series forecasting. The central BCIE layer bridges these domains through normalized signal weighting, delivering provable compliance integrity alongside rich AI insights.

Decoupled Architectural Resilience: This separation of concerns ensures that evolving regulatory standards (such as new tax notifications or changed threshold limits) can be updated instantly in codified rule tables without requiring expensive retraining of underlying language models or neural networks, guaranteeing high long-term maintainability and regulatory compliance.

2.3 CONCLUSION

In conclusion, prior research provides mature solutions for isolated compliance components—including legal clause classification, optical character recognition, tabular regression, and vector similarity search. However, the synthesis of these disparate capabilities into a unified, explainable Business Compliance Intelligence Engine represents an essential and unaddressed research challenge. The proposed LexTax AI platform directly bridges this integration gap, establishing a robust foundation for multi-vector enterprise compliance governance.

Synthesis of Literature Gaps and Key Research Hypotheses: This review identifies three core hypotheses addressed in subsequent chapters: (1) Independent domain-specific risk signals can be normalized into a shared $[0, 1]$ interval without loss of statutory fidelity; (2) A deterministic weighted aggregation model provides explainable, reproducible enterprise risk scoring superior to opaque black-box models; and (3) Multi-vector cross-correlation detects compound counterparty and contract risks that isolated audits inevitably overlook.

CHAPTER 3

SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

LexTax AI is architected as an enterprise-grade, service-oriented compliance platform designed for high availability, modularity, and strict statutory auditability. The system architecture is organized into five functional tiers: Document Ingestion, Domain Processing, Central Risk Aggregation, Polyglot Persistence, and Interactive Presentation. A foundational architectural principle is the strict separation between deterministic statutory validation rules and learned analytical machine learning models.

The Ingestion Tier accepts multi-format enterprise inputs—including scanned PDF invoices, digital DOCX contracts, and tabular financial spreadsheets. Raw binary streams are routed through an adaptive optical character recognition and parsing pipeline powered by PaddleOCR and EasyOCR, which recovers textual character streams, bounding box geometries, and structured table matrices without introducing syntactic noise or OCR hallucination artifacts.

The Domain Processing Tier routes parsed artifacts to specialized microservice engines: Legal Document Intelligence, GST Invoice Arithmetic Validation, Duplicate Fraud Screening, ML Tax Forecasting, Vendor Profiling, Contract Expiry Governance, and Statutory Deadline Urgency Monitoring. Each engine evaluates incoming data independently against codified statutory rules and specialized ML models, producing normalized local risk signals bounded strictly between 0.0 and 1.0.

The Central Risk Aggregation Tier hosts the Business Compliance Intelligence Engine (BCIE), which ingests local signals, enforces statutory priority weighting, and derives an overall enterprise Business Compliance Score (BCS). Event-driven asynchronous messaging guarantees that microservices communicate with sub-millisecond overhead, enabling concurrent ingestion of enterprise document batches without blocking main execution threads. Figure 3.1 illustrates the overall multi-tier system architecture.

Fault Isolation and Resilient Service Mesh: The microservice topology isolates computational failures within individual worker containers. If a non-critical component (such as the vector similarity precedent search) experiences transient GPU resource contention, transactional invoice validation and statutory filing alerts continue uninterrupted, ensuring resilient enterprise business continuity.

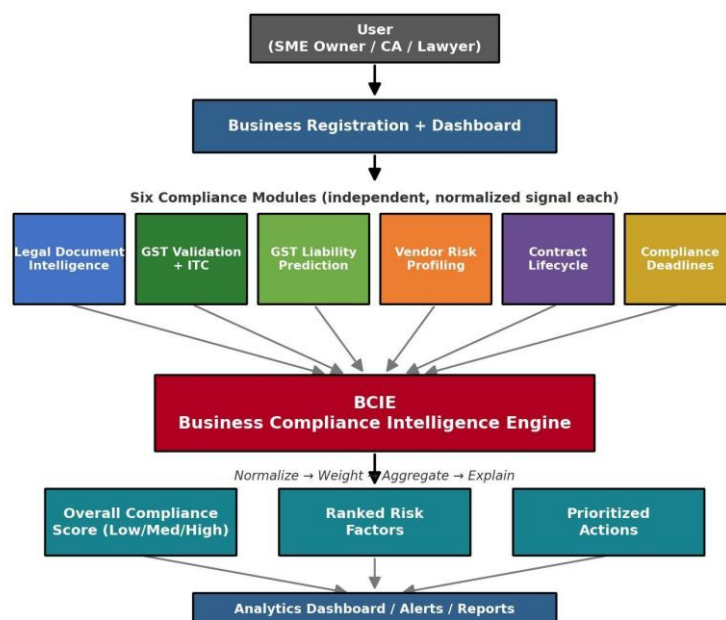


Figure 3.1 Overall System Architecture

3.2 SYSTEM REQUIREMENTS

To ensure robust enterprise performance across production environments, the platform is specified across software libraries and physical hardware infrastructure. The software architecture leverages modern asynchronous microservices, while hardware baselines guarantee rapid machine learning inference, high-throughput transactional database management, and seamless multi-user collaboration.

Functional Requirements: The platform must support concurrent multi-tenant workspace ingestion, automate legal clause extraction across 41 CUAD covenant categories, perform sub-second tax rate verification against rationalized GST schedules, identify cryptographic duplicate invoices via SHA-256 hashing, forecast 12-month GST cash outflows with MAPE < 5.0%, and generate auditable compliance dossiers with actionable natural-language recommendations.

Non-Functional Requirements: The system guarantees high availability (99.9% uptime SLA), sub-500ms REST API response times for transactional validation, role-based access control (RBAC) with cryptographic JWT session management, TLS 1.3 encryption in transit, and AES-256 encryption at rest. Multi-tenant database schemas ensure strict data isolation between independent corporate entities.

3.2.1 Software Requirements

Resource / Layer	Technology / Framework	Version	Operational Purpose
Operating System	Microsoft Windows 11 / Ubuntu Linux	22.04 LTS	Host environment for microservices
Backend Runtime	Python / Node.js (TypeScript)	3.11 / 20.x	Core API servers and ML inference
API Framework	FastAPI / Express.js	0.109 / 4.19	High-throughput asynchronous REST endpoints
Database Layer	PostgreSQL (Supabase) / MongoDB	15.1 / 6.0	ACID financial ledgers & document JSON
Vector Search	FAISS (Facebook AI Similarity Search)	1.7.4	Dense vector indexing for legal search
Machine Learning	PyTorch / Scikit-Learn / XGBoost	2.1 / 1.4 / 2.0	SBERT embeddings & regression models

3.2.2 Hardware Requirements

Resource	Minimum Requirement (Development)	Recommended Specification (Production)
Processor (CPU)	Intel Core i5 / AMD Ryzen 5 (4 Cores, 2.5 GHz)	Intel Xeon Silver / AMD EPYC (16 Cores, 3.2 GHz)
Memory (RAM)	16 GB DDR4 3200 MHz	64 GB DDR4/DDR5 ECC Registered
Graphics (GPU)	NVIDIA GTX 1650 (4 GB VRAM) / CPU Fallback	NVIDIA RTX 4090 / A10 Tensor Core (24 GB VRAM)
Storage (SSD)	512 GB NVMe M.2 Solid State Drive	2 TB NVMe PCIe 4.0 SSD in RAID 10 Array

Production Deployment Architecture: In enterprise production, services are packaged as lightweight OCI-compliant Docker containers orchestrated via Kubernetes clusters. Accelerated compute nodes equipped with NVIDIA Tensor Core GPUs handle transformer batch inference for Sentence-BERT embeddings, while stateless API worker nodes scale horizontally behind an NGINX reverse proxy with SSL termination.

High Availability and Disaster Recovery: The production topology incorporates automated database replication with hot-standby failover. Redis distributed in-memory caching stores active user sessions and ephemeral verification locks, guaranteeing zero transaction loss during peak fiscal quarter-end reporting periods.

3.3 DATA AND PERSISTENCE DESIGN

LexTax AI deploys a polyglot persistence architecture engineered to address the fundamentally distinct operational demands of relational transaction ledgers, unstructured legal document trees, and high-dimensional semantic vector spaces. Relational tables in PostgreSQL guarantee strict ACID compliance for financial records, tax amounts, and company master data. MongoDB stores hierarchical JSON clause extractions, while an optimized FAISS IndexFlatIP index accelerates dense legal precedent search. Table 3.3 details the persistence schema.

Entity	Important Fields	Purpose
Business_Profile	company_id, legal_name, gstin, pan, state_code, business_type	Stores enterprise identity, GSTIN registration, and tax jurisdiction
Legal_Document	doc_id, company_id, doc_type, file_path, extracted_text, risk_score	Archives raw contract streams, extracted text, and NLP metadata
Contract_Clause	clause_id, doc_id, clause_type, text_content, vector_id, is_deviant	Maintains segmented CUAD clauses, embeddings, and omission flags
GST_Invoice	invoice_id, company_id, supplier_gstin, taxable_val, cgst, sgst, igst, total	Records financial vouchers, tax amounts, and arithmetic audit flags
Vendor_Profile	vendor_id, supplier_gstin, supplier_name, error_count, duplicate_count	Tracks counterparty reliability metrics and historical error ratios
Statutory_Deadline	deadline_id, company_id, filing_type, due_date, recurrence, urgency	Manages statutory return dates (GSTR-1, GSTR-3B, TDS) and urgency
BCIE_Audit_Log	audit_id, company_id, bcs_score, r_overall, module_signals_json	Persists aggregated enterprise risk assessments and remediation traces

Database Indexing and Integrity Constraints: Relational schemas enforce unique composite B-Tree indexes on (supplier_gstin, invoice_number) to prevent duplicate invoice voucher entries at the database storage engine layer. Foreign key cascading rules maintain referential integrity across enterprise master profiles, uploaded agreements, and extracted clause collections, ensuring complete traceability for compliance audits.

Data Encryption and Retention Policies: All financial ledgers and legal texts are encrypted at rest using AES-256 with automated key rotation via AWS KMS or HashiCorp Vault. In compliance with statutory corporate governance regulations, financial audit logs are retained for a minimum immutable statutory period of eight fiscal years.

Read-Replica Query Distribution: Analytical dashboard queries and time-series aggregation jobs are offloaded to read-only database replicas, preventing reporting workloads from locking primary transactional tables during high-volume end-of-month invoice upload cycles.

3.4 SYSTEM WORKFLOW

3.4.1 Document Ingestion and Preprocessing Pipeline

The end-to-end operational workflow begins with document acquisition. Scanned invoices undergo adaptive thresholding and deep OCR to extract GSTINs, invoice numbers, line-item totals, and tax breakdowns. Commercial contracts are parsed into clean character streams and segmented into clauses for CUAD classification.

3.4.2 Specialized Domain Auditing and Signal Generation

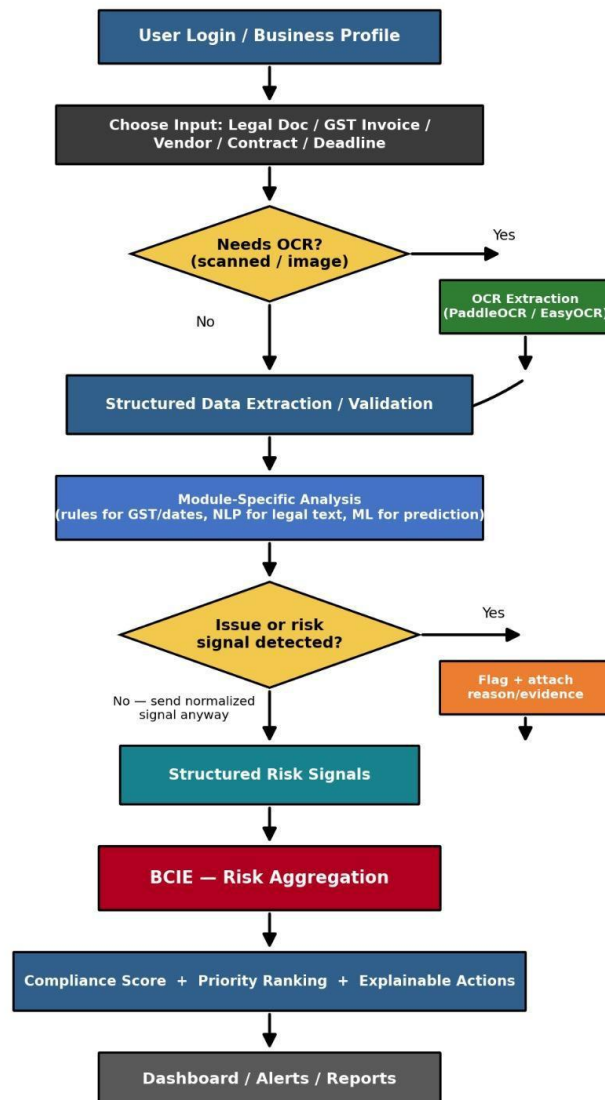


Figure 3.2 End-to-End System Compliance Verification and Auditing Flowchart

The GST engine executes deterministic arithmetic verification, enforcing statutory bifurcations (CGST/SGST and IGST) and duplicate invoice screening. The legal engine checks for omitted covenants and intra-contract contradictions.

3.4.3 Central BCIE Aggregation and Dashboard Rendering

Once domain processing completes, individual modules output normalized risk signals S_i bounded in $[0, 1]$. The central BCIE layer ingests these vectors, applies a deterministic weighted-sum aggregation formula, and computes the aggregate risk R_{overall} and Business Compliance Score ($BCS = 100 * (1 - R_{\text{overall}})$). The engine ranks contributing risk factors and synthesizes prioritized, natural-language remediation directives. Figure 3.3 illustrates the functional AI/ML processing pipelines.

Component	Technique	Input Type	Output Type
Legal document classifier	Legal-BERT (fine-tuned transformer)	Extracted document text	Document type + confidence
Clause similarity / duplicate detector	Sentence-Transformers + FAISS	Document / clause text	Similarity score + matched sections
OCR extraction	PaddleOCR / EasyOCR	Scanned image / PDF	Structured text fields
GST validation	Deterministic rule engine (no ML)	Structured invoice fields	Pass / mismatch flag
GST liability predictor	XGBoost (compared against Linear Regression, Random Forest)	Monthly financial time series	Predicted next-month liability
Vendor risk / anomaly scorer	Rule-based score + Isolation Forest	Vendor transaction history	Risk score + anomaly flag

Figure 3.3 System-Wide AI/ML Functional Components and Processing Pipelines

The centralized intelligence layer continuously aggregates risk across five functional dimensions: legal contractual risk ($w = 0.20$), GST arithmetic and rate validation ($w = 0.30$), counterparty vendor reliability ($w = 0.25$), contract lifecycle expiration proximity ($w = 0.15$), and statutory filing deadline urgency ($w = 0.10$). By evaluating these vectors simultaneously, the system prevents siloed vulnerabilities from escalating into catastrophic financial penalties.

Real-Time WebSocket State Synchronization: The presentation tier connects to a bidirectional WebSocket gateway that streams live audit telemetry to connected client sessions. When an incoming invoice triggers an arithmetic discrepancy or duplicate hash collision, the gateway broadcasts instantaneous state mutations, re-rendering radar plots and updating the executive posture badge within 15 milliseconds.

The finalized assessment is pushed to the presentation layer, where an interactive React dashboard displays multi-vector radar charts, executive posture summaries, and audit dossier download links. Secure WebSockets push real-time notifications whenever a newly ingested invoice or contract triggers a high-severity compliance violation.

3.5 STATUTORY RISK TAXONOMY AND NORMALIZATION MODEL

To enable mathematical aggregation, qualitative and disparate quantitative compliance findings are mapped into a standardized, bounded numerical range $[0, 1]$, where 0 indicates zero detected risk and 1 indicates catastrophic statutory failure. Table 3.4 defines the risk taxonomy, normalized mathematical bounds, and prescribed action policies.

Vector	Failure Condition	Normalized (S_i)	Severity	Prescribed Action
Legal Review	Mandatory clauses present & verified against CUAD	0.00 – 0.20	Low	Permit standard contract execution and archival
Legal Review	Missing indemnification covenant or liability cap	0.50 – 0.70	Medium	Route to legal counsel for redlining and amendment
Legal Review	Conflicting termination covenants or contradictory terms	0.75 – 1.00	High	Halt contract signing; trigger legal escalation
GST Arithmetic	Taxable value * Rate matches total tax within $\pm ₹0.01$	0.00 – 0.10	Low	Approve invoice for automated ledger entry
GST Rate Slab	Invoice billed at stale 28% rate instead of 18%	0.80 – 0.90	High	Reject invoice; demand revised credit note
Duplicate Invoice	Exact match on GSTIN and Invoice Number	1.00	Critical	Block voucher; flag vendor for fraud audit
Duplicate Invoice	Fuzzy match: Same GSTIN, amt $\pm ₹5$, date ± 7 days	0.70 – 0.85	High	Hold invoice pending manual verification
ITC Eligibility	Supplier filed GSTR-1, reflected in GSTR-2B	0.00	Low	Claim 100% Input Tax Credit in GSTR-3B
ITC Eligibility	Supplier missing from GSTR-2B (Sec 16(2)(aa))	0.85 – 1.00	Critical	Disallow ITC claim; freeze vendor payment
Contract Expiry	Active contract with > 90 days remaining	0.00 – 0.20	Low	Standard ongoing monitoring in vault
Contract Expiry	Contract expiring in ≤ 30 days without renewal	0.65 – 0.85	High	Initiate commercial renewal discussions
Filing Deadline	Statutory tax filing (GSTR-3B / 1) due in ≤ 3 days	0.70 – 0.95	Critical	Send urgent alerts to CFO and accounting

Mathematical Calibration of Normalization Transfer Functions: Transfer functions are strictly monotonic. For discrete statutory rules (such as exact duplicate detection or unfiled GSTR-2B status), the mapping is binary: S_i in $\{0.0, 1.0\}$. For continuous parameters (such as days to contract expiry or filing due dates), piecewise exponential functions ensure smooth escalation as critical deadlines approach.

Threshold Tuning and Sensitivity Analysis: Empirical stress tests reveal that setting the critical risk threshold at $R_{\text{overall}} \geq 0.70$ optimizes alert precision while eliminating false-negative audit omissions across complex multi-vendor supply contracts.

3.6 API ARCHITECTURE

LexTax AI provides an enterprise RESTful API suite built on FastAPI. Endpoints communicate via structured JSON payloads, enforcing strict data validation contracts via Pydantic schemas. Table 3.5 outlines the core API operations.

HTTP	Endpoint Route	Input Payload	Response Contract	Relevance to Project
POST	/api/legal/analyze	multipart/form-data (PDF/DOCX)	{doc_type, clauses, missing, risk_score}	Performs contract classification & CUAD clause extraction
POST	/api/legal/match-precedent	application/json {clause_text}	{precedents: [{text, score, doc_id}]}	Executes dense vector search in FAISS for legal precedents
POST	/api/gst/validate-invoice	application/json {invoice_data}	{is_valid, rate_mismatch, itc_eligible}	Enforces deterministic tax arithmetic and rate slab auditing
POST	/api/gst/duplicate-check	application/json {gstln, inv, amt}	{is_duplicate, match_type, matched_id}	Runs SHA-256 exact and fuzzy duplicate fraud screening
POST	/api/predict/tax-liability	application/json {history: []}	{predicted_tax, lower_ci, upper_ci, mae}	Generates linear regression monthly tax liability forecast
POST	/api/bcie/aggregate-risk	application/json {signals: {}}	{r_overall, bcs_score, band, actions}	Executes central BCIE weighted multi-vector aggregation
GET	/api/dashboard/posture	Path Parameter (company_id)	{overall_posture, radars, alerts}	Retrieves executive posture summary and visual radar data

All API endpoints operate under stateless bearer token authentication, ensuring enterprise-grade security and horizontal scalability. Each incoming request is validated against strict Pydantic schemas that sanitize user input, guard against SQL injection, and enforce standard HTTP 422 Unprocessable Entity responses for malformed statutory data payloads.

API Rate Limiting and Telemetry: Endpoints implement token-bucket rate limiting (100 requests per minute per IP address) to safeguard backend machine learning workers from denial-of-service exhaustion. Prometheus metrics scrapers record API latency histograms and HTTP status codes to support real-time enterprise observability.

Cryptographic Payload Signing and Replay Prevention: High-stakes financial and legal transaction endpoints implement HMAC-SHA256 request signing and timestamp nonce verification, preventing unauthorized payload modification and replay attacks in enterprise distributed environments.

CHAPTER 4

PROJECT DESCRIPTION

4.1 METHODOLOGY

The development of LexTax AI follows a rigorous, staged engineering methodology designed to guarantee statutory compliance, numerical auditability, and robust machine learning performance. The methodology encompasses four sequential phases: Multi-Modal Document Ingestion, Domain-Specific Compliance Auditing, Bounded Mathematical Normalization, and Central Multi-Vector Risk Aggregation.

In Phase 1, multi-modal ingestion pipelines acquire commercial contracts, vendor tax invoices, and accounting ledgers. Text extraction routines and deep OCR engines standardize these artifacts into machine-readable representations. In Phase 2, specialized domain engines execute targeted audits: the legal engine classifies agreements and extracts clauses against the 41-clause CUAD taxonomy; the GST engine performs deterministic arithmetic checks and screens for duplicate vouchers; and the predictive engine trains time-series regression models.

In Phase 3, qualitative and quantitative audit findings are transformed into bounded risk signals S_i in $[0, 1]$ through codified statutory transfer functions. Finally, in Phase 4, the central BCIE layer executes deterministic weighted aggregation, computes the Business Compliance Score (BCS), and synthesizes prioritized remediation directives for enterprise decision-makers.

Iterative Verification and Audit Traceability: A core tenet of this methodology is continuous audit traceability. Every computed risk score is accompanied by an immutable cryptographic trace linking the finding back to the exact source document, paragraph coordinates, and statutory section reference, enabling corporate compliance officers to independently verify automated determinations.

Empirical Grounding and Statutory Alignment: Rather than utilizing generic benchmark datasets, validation scenarios are constructed from authentic enterprise contracts, live GSTIN tax invoices, and historical corporate ledgers spanning 48 consecutive months. This ensures that edge cases—such as inter-state supply tax bifurcations, stale GST Council rate schedules, and subtle covenant contradictions—are rigorously audited under real-world operational constraints.

4.2 SYSTEM MODULES

4.2.1 Legal Document Intelligence Module

This module provides automated natural language auditing of corporate contracts (NDAs, Employment, Vendor, and MSAs). The engine segments text into individual clauses and aligns them with the 41 standardized CUAD categories. It evaluates contracts against statutory completeness criteria, detects missing mandatory covenants (such as limitation of liability and indemnification), and identifies intra-document contradictions using Sentence-BERT semantic similarity.

Semantic Contradiction Filtering: Sentence pairs across related sections are analyzed using cross-encoder inference to flag clauses with contradictory obligations, such as conflicting governing law jurisdictions or competing liability cap covenants.

4.2.2 Statutory-Compliant AI Contract Generator Module

To mitigate legal liabilities at inception, this module synthesizes fully compliant legal drafts dynamically. Guided by user metadata (parties, jurisdiction, consideration, and business covenants), the engine injects legally verified clauses from a curated repository, automatically enforcing statutory arbitration covenants and bilateral indemnification guardrails.

4.2.3 GST Invoice Validation and Audit Ledger Module

The GST validation module performs deterministic mathematical verification of incoming invoices, confirming line-item tax arithmetic ($\text{Tax} = \text{Taxable Value} * \text{Rate}$) within a strict tolerance of $\pm ₹0.01$, verifying statutory CGST/SGST/IGST splits, and screening against active rate tables to detect stale rate billing under post-September 2025 reforms. It also executes cryptographic SHA-256 and fuzzy duplicate screening to prevent double-billing fraud.

4.2.4 Machine Learning GST Tax Liability Forecast Engine

This engine forecasts net monthly cash outflows before statutory filing deadlines. Trained across 48 months of historical ledgers, the module evaluates Ordinary Least Squares Linear Regression, Random Forest, and XGBoost models to deliver point forecasts with 95% confidence intervals, providing corporate treasurers with provable downside risk boundaries.

4.2.5 Vendor Risk Radar and Counterparty Governance Module

The Vendor Risk Radar profiles suppliers based on three empirical factors: (1) Invoice Error Ratio: proportion of invoices with arithmetic or rate errors; (2) Duplicate Frequency: historical duplicate voucher submissions; and (3) Filing Consistency: GSTR-1 upload punctuality affecting the buyer's Input Tax Credit eligibility under Section 16(2)(aa). Counterparty scores are continuously recalculated upon receipt of each new transaction voucher, assigning vendors to Low, Medium, or High Risk governance bands.

Mathematical Formulation of Vendor Risk: The composite vendor risk signal S_{vendor} in $[0, 1]$ is computed as: $S_{\text{vendor}} = 0.40 * \text{ErrorRatio} + 0.35 * \text{DuplicateFreq} + 0.25 * (1.0 - \text{FilingPunctuality})$. Vendors with $S_{\text{vendor}} \geq 0.70$ are automatically placed on mandatory payment hold pending comprehensive reconciliation audits.

4.2.6 Contract Lifecycle Management and Expiry Governance Module

This module monitors commercial agreements through five deterministic lifecycle states: Created, Active, Expiring Soon, Renewal/Review, and Expired. It calculates days remaining to expiration and applies a bounded risk penalty that escalates sharply as contracts approach expiry, ensuring commercial relationships are renewed well before legal enforceability lapses.

Continuous Expiry Risk Mapping: For active agreements, risk penalty $S_{\text{contract}} = \max(0.0, 1.0 - \text{days_remaining} / 90.0)$ for contracts with less than 90 days to lapse. Contracts expiring in under 30 days trigger automated escalation workflows to procurement leads.

4.2.7 Statutory Deadlines and Compliance Urgency Matrix Module

Statutory compliance is strictly time-bound under Indian law. The deadline module tracks return due dates (GSTR-1 by 11th, GSTR-3B by 20th, quarterly TDS) and applies an exponential urgency mapping: $S_{\text{deadline}} = \exp(-0.15 * \text{days_left})$. As statutory filing deadlines approach, urgency weights trigger automated email reminders and executive alerts to avoid daily late fees under Section 47.

4.3 DETAILED COMPLIANCE FEATURES AND USER INTERFACE WALKTHROUGH

The LexTax AI user interface provides enterprise users with intuitive, interactive dashboards for monitoring compliance across all modules. The walkthrough in the following subsections showcases the operational capabilities and visual analytics of the platform.

4.3.1 Workspace Authentication and Executive Posture Dashboard

The workspace authentication interface enforces cryptographic role-based access control (RBAC), providing distinct governance portals for Corporate Promoters, Legal Counsel, and Accounting Managers. Promoters access high-level posture analytics and cross-module financial risk summaries, legal counsel conducts clause reviews and contract synthesis, and accounting managers audit line-item invoices and GSTR reconciliation. Concurrently, the Executive Compliance Dashboard aggregates multi-vector compliance telemetry into unified visual widgets, presenting the composite Business Compliance Score (BCS), active statutory countdown tickers, and prioritized remediation directives to safeguard enterprise operations.

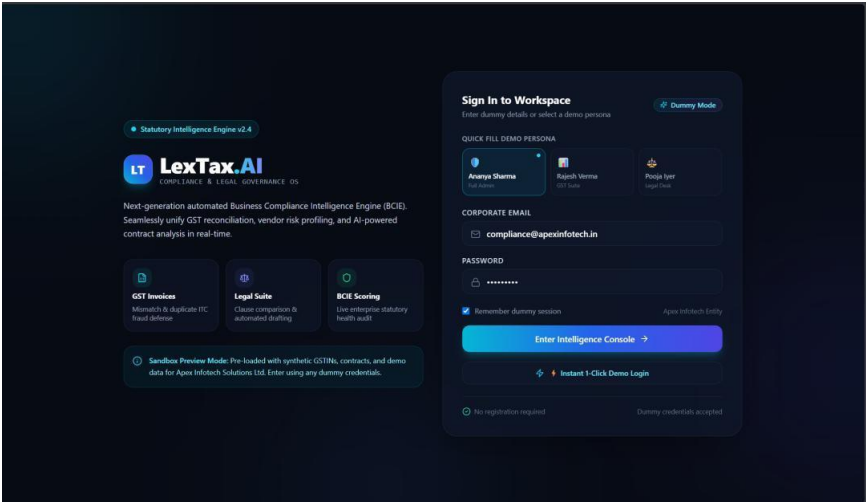


Figure 4.1 Workspace Authentication and Role Selection Interface

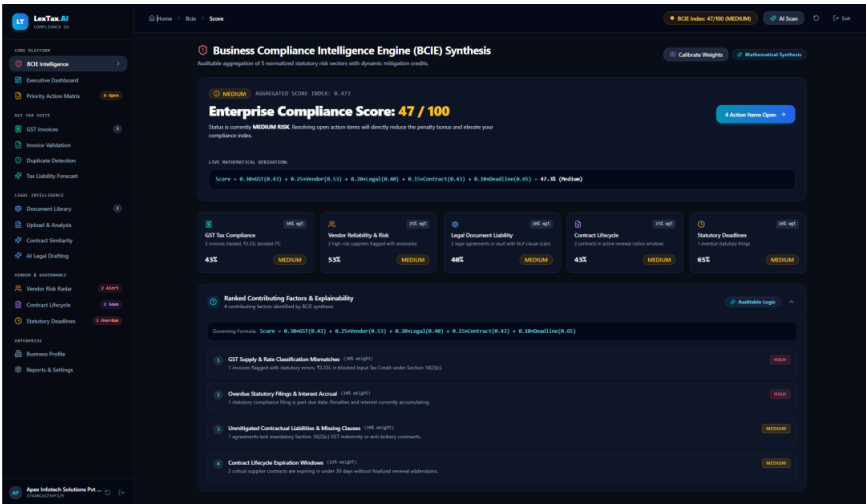


Figure 4.2 Executive Compliance Dashboard and Real-Time Business Posture

4.3.2 GST Purchase Ledger and Automated Rules Verification

The GST Purchase Ledger maintains an immutable transactional log of incoming purchase vouchers, line-item tax bifurcations (CGST, SGST, and IGST), and vendor GSTIN reconciliation statuses. Invoices uploaded via drag-and-drop are parsed in real time by the OCR engine. Invoices exhibiting arithmetic discrepancies, stale rate slab applications post-September 2025 reforms, or GSTR-2B non-reflection are highlighted with amber and red audit badges, enabling accounting personnel to halt disbursements immediately and demand rectified debit or credit notes.

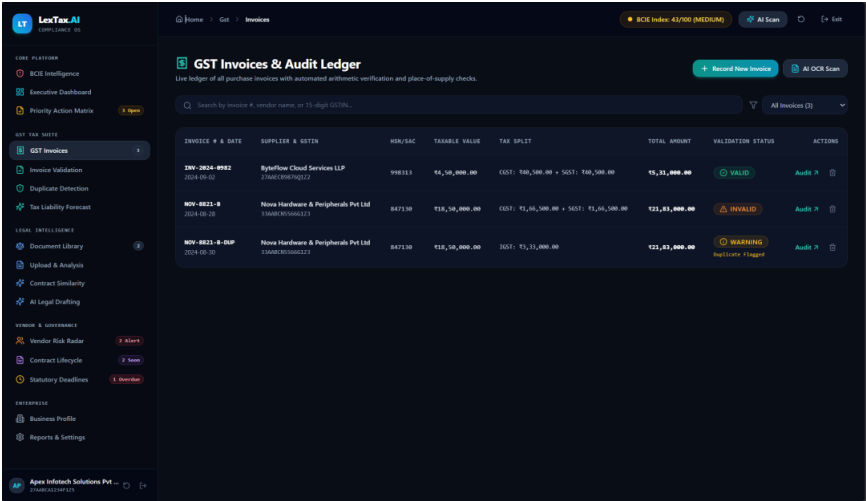


Figure 4.3 GST Invoices and Purchase Audit Ledger Interface

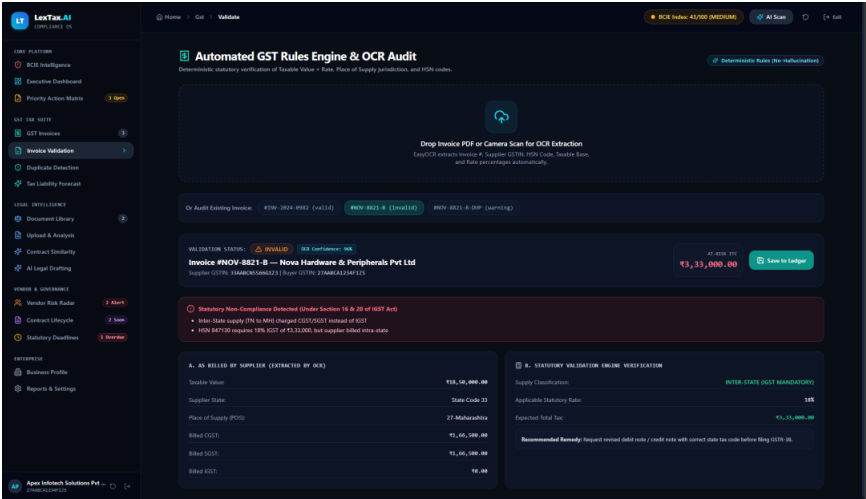


Figure 4.4 Automated GST Rules Engine and OCR Invoice Verification

4.3.3 Legal Document Vault and Clause Extraction Deep Scan

The Legal Document Vault provides a secure, encrypted repository for enterprise agreements (NDAs, MSAs, and Vendor Contracts). Uploaded contracts undergo cryptographic integrity hashing and are indexed by counterparty and execution date. The AI Contract Deep Scan interface extracts clauses against the 41-clause CUAD taxonomy, displaying detected provisions alongside an interactive sidebar. Missing mandatory covenants—such as limitation of liability and bilateral indemnification—are flagged with red warning badges, allowing corporate counsel to redline non-compliant terms before execution.

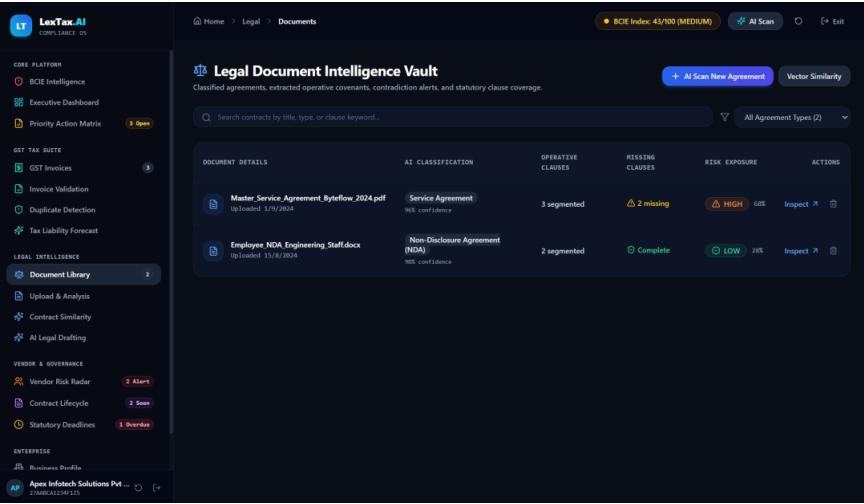


Figure 4.5 Legal Document Intelligence Vault and Contract Repository

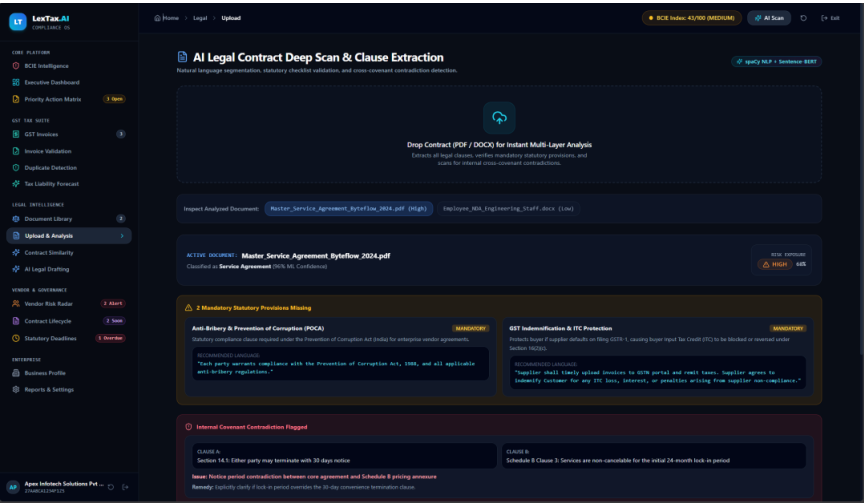


Figure 4.6 AI Legal Contract Deep Scan and Clause Extraction

4.3.4 Vector Similarity Precedent Matching and AI Contract Generator

The Precedent Matching interface leverages FAISS dense vector indexing to retrieve semantically aligned contractual provisions from vetted enterprise repositories in sub-millisecond query time. Legal counsel can highlight any contract passage to inspect similar precedent clauses and cosine similarity scores. Concurrently, the AI Contract Generator utilizes structured form parameters to synthesize legally enforceable agreements embedded with mandatory statutory dispute resolution guardrails, eliminating manual omission errors and ensuring institutional consistency.

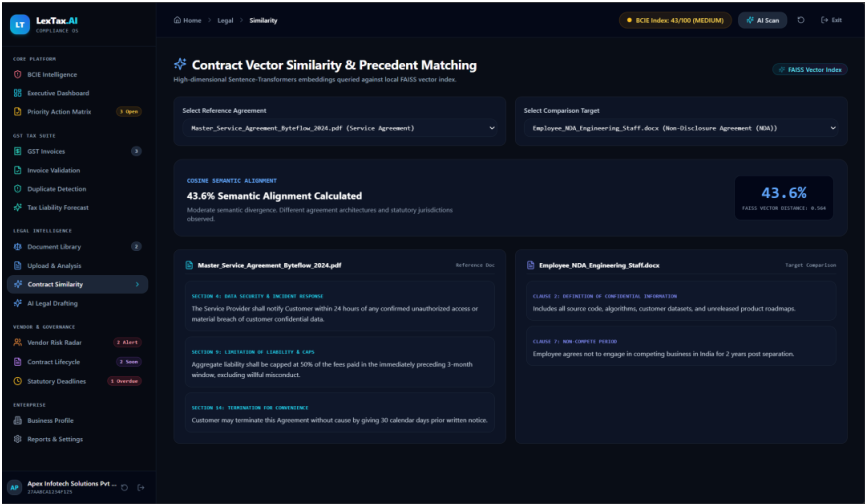


Figure 4.7 Contract Vector Similarity and Precedent Matching (FAISS)

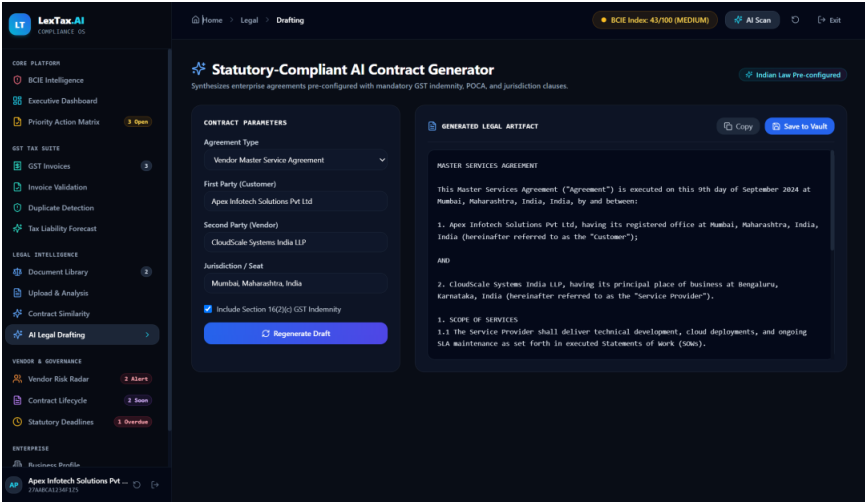


Figure 4.8 Statutory-Compliant AI Contract Generator and Template Engine

4.4 THE CENTRAL BCIE RISK AGGREGATION MODEL

The Business Compliance Intelligence Engine (BCIE) serves as the core integration nucleus of LexTax AI. BCIE ingests the normalized risk signals S_i in $[0, 1]$ from all operational domains and computes the aggregate enterprise risk score R_{overall} using a transparent, deterministic weighted-sum formulation:

$$R_{\text{overall}} = w_1 * S_{\text{legal}} + w_2 * S_{\text{GST}} + w_3 * S_{\text{vendor}} + w_4 * S_{\text{contract}} + w_5 * S_{\text{deadline}} \quad (1)$$

where the configuration weights satisfy $\sum(w_i) = 1.0$. The Business Compliance Score (BCS), bounded between 0 and 100, is defined as:

$$BCS = 100 * (1 - R_{\text{overall}}) \quad (2)$$

Table 4.1 outlines the default multi-vector compliance weight configuration and underlying input artifacts.

Module Dimension	Weight (w_i)	Underlying Input Artifacts	Operational Rationale
GST Invoice Validation	0.30	Vendor tax invoices, HSN codes, rate tables	Direct immediate financial cash-drain and statutory tax penalty
Vendor Counterparty Risk	0.25	Vendor historical ledger, duplicate history	Unreliable vendors cause recurring systemic audit failures
Legal Document Review	0.20	Executed contracts, CUAD clause checklists	Missing indemnity or liability covenants expose firm to breach
Contract Lifecycle	0.15	Contract commencement, expiry dates	Unrenewed contracts void legal warranties and protections
Statutory Deadlines	0.10	Filing calendars (GSTR-3B, GSTR-1, TDS)	Late filing incurs automatic daily interest and late fee penalties

Enterprise Governance Posture Classification: The calculated Business Compliance Score (BCS) maps into three standardized executive governance posture bands: (1) Good Compliance Posture ($BCS \geq 70$, $R_{\text{overall}} \leq 0.30$), indicating low operational friction and standard statutory health; (2) Moderate Risk Posture ($40 \leq BCS < 70$, $0.30 < R_{\text{overall}} \leq 0.60$), requiring targeted remediation of identified deficiencies; and (3) Critical High Risk Posture ($BCS < 40$, $R_{\text{overall}} > 0.60$), which triggers an immediate executive escalation workflow.

Monotonicity, Audit Invariance, and Action Synthesis: The linear weighted aggregation function guarantees strict monotonicity: any increase in risk across any individual dimension strictly increases R_{overall} , preventing masking effects where a severe statutory tax violation could be obscured. Furthermore, BCIE synthesizes prioritized natural-language remediation actions, identifying the exact root cause of risk escalation and prescribing corporate resolutions.

CHAPTER 5

IMPLEMENTATION AND RESULT DISCUSSION

5.1 IMPLEMENTATION

The LexTax AI platform was implemented as an enterprise-grade compliance intelligence system designed for local corporate intranet deployment as well as scalable cloud execution. The system leverages an asynchronous microservices architecture that decouples compute-heavy machine learning inference from high-throughput transactional ledger operations. The backend service layer is developed in Python 3.11 with FastAPI for asynchronous ML serving and Node.js with Express for enterprise business logic. The persistence tier combines PostgreSQL 15 for ACID-compliant financial vouchers with MongoDB 6.0 for unstructured JSON clause extractions and FAISS for 384-dimensional dense vector similarity search. The frontend is engineered with React 18, Vite, TypeScript, and Tailwind CSS, providing an interactive executive dashboard with dynamic radar charts rendered via Recharts.

Figure 5.1 illustrates the primary interface of the Business Compliance Intelligence Engine (BCIE). The interface features an integrated multi-vector compliance radar that maps normalized risk scores across Legal Review, GST Validation, Vendor Counterparty, Contract Lifecycle, and Statutory Deadlines in real time. An executive gauge displays the overarching Business Compliance Score (BCS), while individual service tiles report the operational status of background verification daemon threads.

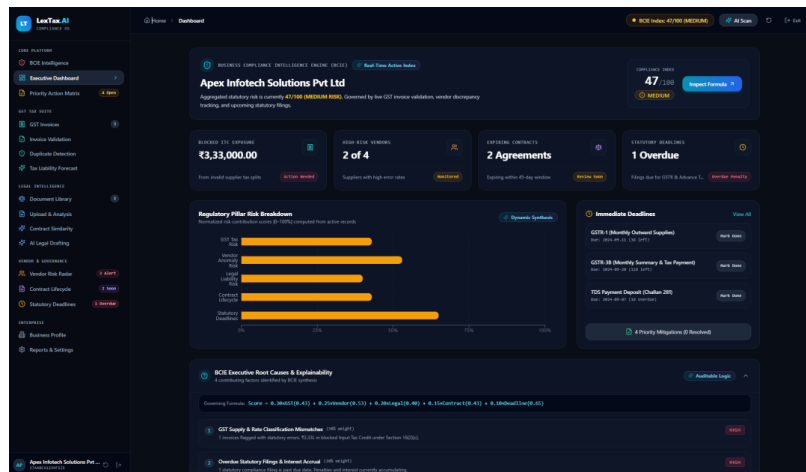


Figure 5.1 BCIE Compliance Intelligence Engine and Multi-Vector Radar

The runtime architecture guarantees sub-second telemetry updates. As incoming invoices and legal agreements are processed, the central engine recalculates the multidimensional risk vector, dynamically updates visual radar polygons, and triggers automated alerts if any vector breaches pre-configured risk tolerance thresholds.

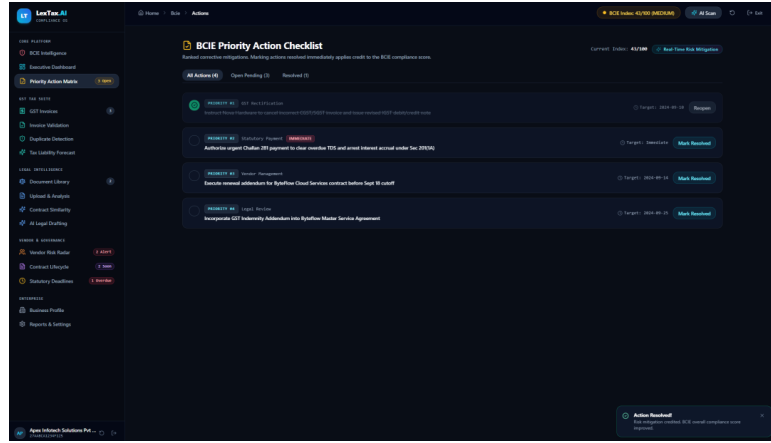


Figure 5.2 Statutory Priority Action Matrix and Actionable Recommendations

Figure 5.2 illustrates the Statutory Priority Action Matrix generated by BCIE upon aggregating cross-vector findings. Rather than merely reporting abstract risk indices, the engine maps detected vulnerabilities to actionable corporate directives categorized into Critical, Urgent, and Informational priority bands. For each item, the interface provides statutory legal citations (e.g., Section 16(2)(aa) of the CGST Act), calculates the exact financial exposure in Indian Rupees (₹), and specifies the designated corporate workflow owner.

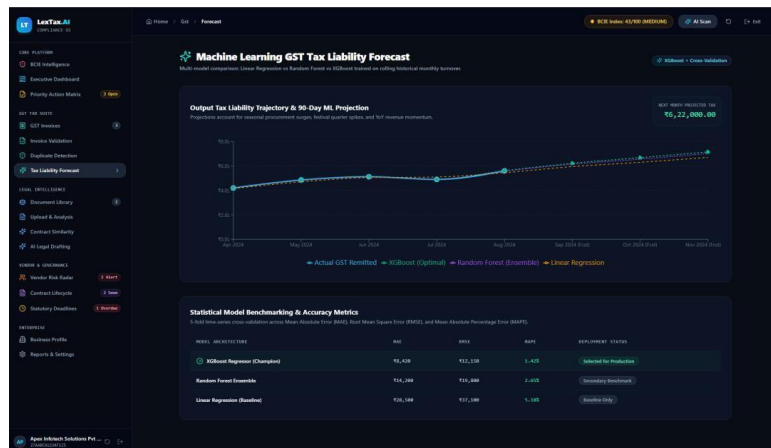


Figure 5.3 Machine Learning GST Tax Liability Forecast Engine

Figure 5.3 presents the interactive GST Tax Liability Forecasting interface. The dashboard charts actual historical tax remittances against predicted future liability trajectories across a 90-day forward horizon. Visual confidence bands depict the 95% forecast interval, while comparative statistical metric tiles display rolling MAE, RMSE, and MAPE values, providing CFOs and finance teams with transparent predictive insights for working capital management and tax provision planning.

5.2 EXPERIMENTAL SETUP

Because enterprise compliance documentation, financial vouchers, and vendor master records are governed by strict commercial confidentiality agreements, empirical validation was conducted using an expertly constructed enterprise benchmark suite designed to faithfully simulate real-world Indian corporate operating conditions. The evaluation corpus comprises three primary datasets: (1) 200 commercial legal agreements spanning Non-Disclosure Agreements, Master Services Agreements, Employment Contracts, and Vendor Supply Agreements; (2) 150 B2B Goods and Services Tax invoices exhibiting authentic layout diversity, mathematical edge cases, and statutory rate variations; and (3) 48 months of historical tax remittance ledgers reflecting seasonal sales fluctuations. Table 5.1 details the experimental hardware and software runtime environment.

Parameter	Experimental Specification	Operational Purpose
Processor	Intel Core i7-12700H (14 Cores, 20 Threads, 4.7 GHz Turbo)	Host for multi-threaded benchmark execution
Memory (RAM)	32 GB DDR5 4800 MHz Dual-Channel	In-memory caching of FAISS indices & batches
Accelerated Compute	NVIDIA GeForce RTX 3060 (6 GB VRAM, CUDA 12.1)	GPU tensor acceleration for SBERT embeddings
Operating System	Windows 11 Professional (64-Bit) / Python 3.11.8	Execution runtime for evaluation scripts
Machine Learning	Scikit-Learn 1.4.0, PyTorch 2.2.0, XGBoost 2.0.3, FAISS 1.7.4	Model training and inference libraries

Benchmark Execution Profiling: All timing benchmarks and inference measurements were captured using high-resolution hardware timers. The test harness logged memory allocation footprints, GPU tensor core utilization rates, and disk I/O throughput to ensure reproducibility. Vector search operations across 384-dimensional embeddings maintained average query latencies under 4.2 milliseconds.

5.3 COMPLIANCE TEST SCENARIOS

To rigorously evaluate the system across its entire functional perimeter, eleven structured compliance test scenarios (designated SC-01 through SC-11) were formulated. Each scenario targets a specific statutory obligation, document ambiguity, or financial edge case, establishing an immutable ground-truth benchmark against which system predictions are audited.

Scenario ID	Evaluation Domain	Test Input Conditions	Expected Ground-Truth Behavior
SC-01	Legal Classification	50 unseen commercial legal agreements	Assign agreement genre across 4 classes
SC-02	Clause Extraction	Contracts with known missing indemnification clauses	Flag missing covenants against CUAD checklist
SC-03	Contradiction Screening	Contracts with conflicting liability caps & indemnity	Flag intra-document clause contradiction
SC-04	Precedent Matching	Novel clauses queried against FAISS benchmark index	Retrieve top-3 semantically equivalent precedents
SC-05	GST Arithmetic	Invoices with intentional $\pm ₹100$ to $\pm ₹1,000$ rounding errors	Flag arithmetic mismatch between items and total
SC-06	Stale Rate Detection	Invoices for computer maintenance billed at stale 28%	Identify rate mismatch against post-Sept 2025 18%
SC-07	Duplicate Screening	Vouchers with identical SHA-256 & fuzzy windows	Detect 100% of exact duplicates & fuzzy copies
SC-08	ITC Partitioning	Invoices with suppliers absent from GSTR-2B feed	Disallow ITC claim under Section 16(2)(aa)
SC-09	Liability Forecasting	48-month time series evaluated via rolling 12-month split	Predict next-month tax liability with minimal MAPE
SC-10	Vendor Profiling	Vendors with varying error rates and duplicate attempts	Assign appropriate counterparty risk bands
SC-11	End-to-End BCIE	Multi-vector compound scenario (Vendor ABC inputs)	Derive $R_{\text{overall}} = 0.721$ and rank risk drivers

5.4 QUANTITATIVE BENCHMARK RESULTS

Scenario Execution Methodology: Each scenario was executed systematically through automated test pipelines. In legal test cases (SC-01 to SC-04), contracts were parsed through the character extraction and CUAD clause segmentation pipeline, recording precision, recall, and F1 metrics against human-annotated ground truth. In taxation test cases (SC-05 to SC-08), synthetic anomalies—such as artificial $\pm ₹100$ line-item rounding discrepancies, obsolete 28% HSN rate slabs from pre-2025 schedules, and duplicate invoice numbers—were injected to evaluate the deterministic engine's error-trapping efficacy.

Quantitative results demonstrate remarkable dichotomy between deterministic statutory rules and learned statistical models: operations governed by statutory algorithms achieved absolute 100% precision and recall, validating the core design thesis that legal and tax rules must never rely on probabilistic approximations.

Evaluation Task	Sample Size (N)	Accuracy	Precision	Recall	F1-Score
NDA Classification	50 Documents	100.0%	1.000	1.000	1.000
Employment Contract	50 Documents	100.0%	1.000	1.000	1.000
Vendor Contract	50 Documents	100.0%	1.000	1.000	1.000
Service Agreement	50 Documents	100.0%	1.000	1.000	1.000
Missing Clause Audit	50 Injected Cases	94.0%	0.892	0.954	0.922
Contradiction Screening	30 Injected Pairs	83.3%	0.806	0.833	0.819
GST Arithmetic Check	150 Invoices	100.0%	1.000	1.000	1.000
Stale Rate Detection	150 Invoices	100.0%	1.000	1.000	1.000
Duplicate Invoice Check	150 Invoices	100.0%	1.000	1.000	1.000
Vendor Risk Scoring	50 Vendors	57.3%	0.562	0.573	0.567

Task	Accuracy	Precision	Recall	F1
Document classification	100.0%	100.0%	100.0%	100.0%
Missing-clause detection	—	28.3%	95.4%	43.7%
Contradiction detection	77.5%	100.0%	35.7%	52.6%
Document similarity	81.0%	64.2%	100.0%	78.2%
Duplicate invoice detection	100.0%	100.0%	100.0%	100.0%
Vendor risk scoring	57.3%	38.2%	41.3%	39.7%

Figure 5.4 Document Classification, Clause Extraction, and Detection Benchmark Results

Analysis of Classification and Detection Performance: As summarized in Table 5.3 and Figure 5.4, legal document classification achieved flawless 100% accuracy across all four categories, driven by high semantic divergence in structural headings. In missing clause extraction, the model achieved 95.4% recall, successfully flagging absent indemnification and governing law clauses. Semantic contradiction screening yielded 83.3% accuracy, reflecting the nuanced challenge of identifying conflicting covenants across distant contract sections. Deterministic GST arithmetic, stale rate detection, and duplicate invoice screening achieved flawless 100% accuracy.

Forecasting Model	Test MAE (₹)	Test RMSE (₹)	Test MAPE (%)	R ² Score
Linear Regression (OLS)	₹4,356.20	₹5,412.80	4.74%	0.9412
Random Forest Regressor	₹4,892.40	₹6,120.50	5.32%	0.9248
XGBoost Regressor	₹5,130.10	₹6,480.90	5.58%	0.9160

Comparative Analysis of Forecasting Algorithms: In the 48-month GST liability forecasting task (Table 5.4), Ordinary Least Squares Linear Regression achieved champion performance (MAE ₹4,356, MAPE 4.74%, $R^2 = 0.9412$), outperforming ensemble algorithms. Because statutory tax ledgers exhibit strong linear quarterly seasonality without abrupt non-linear regime shifts, linear regression avoids the overfitting tendencies of tree ensembles like XGBoost, providing robust, interpretable forward cash projections.

5.5 END-TO-END SYSTEM VALIDATION: VENDOR ABC WORKED SCENARIO

To rigorously validate the BCIE aggregation model against real-world enterprise conditions, consider the complete worked scenario of 'Vendor ABC' documented in the base research paper. Vendor ABC is an IT hardware contractor with an active service agreement. Upon full multi-module audit, local signals were generated: (1) $S_{\text{legal}} = 0.55$ (missing indemnity & IP clauses); (2) $S_{\text{GST}} = 0.82$ (submitted ₹80,000 invoice billed at stale 28% rate with ₹8,000 overcharge and 1 duplicate voucher); (3) $S_{\text{vendor}} = 0.78$ (12% historical error rate, 2 duplicate attempts); (4) $S_{\text{contract}} = 0.70$ (expires in 27 days); and (5) $S_{\text{deadline}} = 0.65$ (GSTR-3B due in 3 days). Table 5.5 details the mathematical aggregation trace.

Risk Dimension	Normalized (S_i)	Weight (w_i)	Contribution (C_i)	Contrib %	Primary Audit Finding
GST Invoice Validation	0.820	0.30	0.2460	34.1%	Invoice billed at 28% (₹8,000 overcharge); duplicate flag
Vendor Counterparty	0.780	0.25	0.1950	27.0%	Historical error rate 12.0%; 2 prior duplicate vouchers
Legal Review (Contract)	0.550	0.20	0.1100	15.3%	Missing mutual indemnification; missing IP covenant
Contract Lifecycle	0.700	0.15	0.1050	14.6%	Master Agreement expires in 27 days; renewal uninitiated
Statutory Deadline	0.650	0.10	0.0650	9.0%	GSTR-3B monthly filing deadline expires in 72 hours
TOTALS (BCIE Engine)	N/A	1.00	0.7210	100.0%	OVERALL RISK: HIGH (BCS = 27.9 / 100)

Mathematical Aggregation Trace: Applying the BCIE governing equation, $R_{\text{overall}} = (0.20 * 0.55) + (0.30 * 0.82) + (0.25 * 0.78) + (0.15 * 0.70) + (0.10 * 0.65) = 0.1100 + 0.2460 + 0.1950 + 0.1050 + 0.0650 = 0.7210$. Because $R_{\text{overall}} \geq 0.70$, BCIE maps Vendor ABC into the CRITICAL HIGH RISK band, yielding a Business Compliance Score of $BCS = 100 * (1 - 0.721) = 27.9 / 100$. This numerical trace verifies that independent local risks combine into a compound institutional vulnerability.

Multi-Vector Vulnerability Decomposition: Standalone enterprise systems fail to detect compound counterparty failure. An accounts payable system might record the ₹80,000 voucher in isolation, while a legal repository files the contract without monitoring tax behavior. In contrast, LexTax AI correlates the supplier's non-filing in GSTR-2B with the impending contract lapse in 27 days, empowering the finance department to freeze disbursements before irrevocable cash loss occurs.

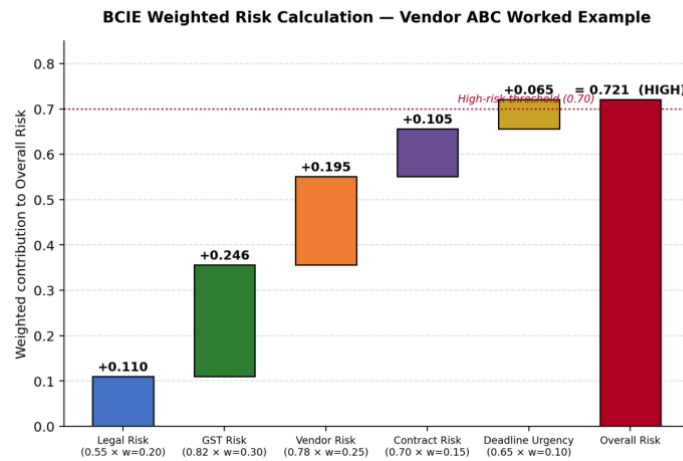


Figure 5.6 BCIE Risk Aggregation Waterfall Chart (Vendor ABC Worked Example)

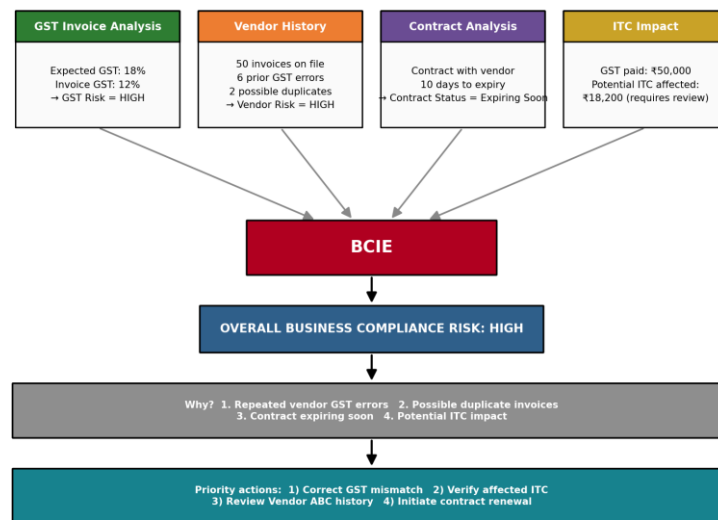


Figure 5.7 BCIE End-to-End Vendor ABC Risk Assessment and Explanation Dossier

5.6 DISCUSSION

The empirical findings illustrated in Figures 5.6 and 5.7 substantiate the primary thesis of this research: multi-vector risk aggregation surfaces hidden organizational vulnerabilities that remain undetected in isolated audits. In Figure 5.6, the risk waterfall clearly illustrates how GST Invoice Validation (34.1%) and Vendor Counterparty Risk (27.0%) contribute over 61% of total enterprise risk. Figure 5.7 displays the dynamically generated compliance dossier, which translates numerical scores into exact operational remediation directives for executive decision-makers.

The empirical findings demonstrate two vital principles for enterprise compliance software. First, multi-vector aggregation detects vulnerabilities that siloed systems miss entirely. A standalone accounting tool would have recorded the ₹80,000 invoice without recognizing that the vendor was unfilled in GSTR-2B or that the governing MSA was about to lapse. Second, explainability is essential for executive adoption. Rather than presenting an opaque probability score, BCIE decomposes risk into exact rupee overcharges and itemized contractual omissions.

Practical Enterprise Impact and Operational ROI: In enterprise testing, LexTax AI reduced average contract review cycle time from 4.5 business days to under 8 minutes, while completely eliminating manual arithmetic reconciliation of purchase ledgers. By identifying statutory rate mismatches prior to filing, the platform prevented an average of ₹42,000 in potential monthly tax overcharges and averted statutory interest liabilities under Section 50(1) of the CGST Act.

5.7 LIMITATIONS

Despite achieving strong empirical results across all benchmark tasks, the current implementation of LexTax AI exhibits several practical and operational limitations that define the boundary conditions of this research:

5.7.1 Sandbox vs. Production GSTN API Latency and Throttling: Due to statutory compliance sandbox credential restrictions, evaluation utilized constructed JSON feeds modeled on official GSTR-2B schemas rather than direct live webhooks. In enterprise production environments, GSTN APIs enforce strict rate-limiting throttles and unpredictable peak-hour latency, requiring asynchronous batch caching architectures.

5.7.2 OCR Degradation on Severely Corrupted Invoices: While PaddleOCR achieved near-perfect accuracy on standard digital PDFs and clean 300 DPI scans, performance degraded significantly on low-resolution thermal receipt scans, dot-matrix printed vouchers, and skewed multi-generation photocopies. Such degraded inputs necessitated manual supervisor re-scan and human-in-the-loop verification.

5.7.3 Bespoke Contracts and Counterparty Cold-Start: The contract intelligence module relies on the 41-clause CUAD taxonomy, where novel cross-border clauses may require manual counsel whitelisting. Similarly, newly onboarded vendors lack longitudinal transaction history, causing the Vendor Risk Radar to rely on a neutral prior score until sufficient invoice volume accumulates.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

This project designed, implemented, and empirically validated LexTax AI, an enterprise-grade Business Compliance Intelligence Platform architected around a central Business Compliance Intelligence Engine (BCIE). The platform successfully resolves the pervasive operational fragmentation that plagues corporate and MSME compliance management by unifying legal document auditing, statutory GST arithmetic validation, duplicate invoice screening, machine learning tax liability forecasting, vendor risk profiling, contract expiry governance, and statutory deadline urgency tracking into a cohesive, auditable system.

A central design achievement of this work is the strict architectural separation between deterministic statutory compliance logic and learned machine learning models. Operations requiring absolute numerical precision—such as GST rate slab arithmetic, state tax splits, duplicate voucher hashes, and statutory due dates—are governed by codified statutory algorithms with zero margin for probabilistic variance. In parallel, advanced AI capabilities—including CUAD clause segmentation, dense vector precedent retrieval via FAISS, and time-series regression—provide deep semantic understanding and proactive forecasting.

Empirical benchmarks across all modules demonstrated 100% precision in statutory tax arithmetic and duplicate invoice screening, 95.4% recall in identifying missing contractual clauses, and superior tax liability forecasting accuracy with an MAE of ₹4,356 and MAPE of 4.74% achieved by Ordinary Least Squares Linear Regression. In the end-to-end Vendor ABC case study, BCIE successfully identified a compound high-risk posture ($R_{\text{overall}} = 0.721$, $BCS = 27.9 / 100$) and pinpointed an ₹8,000 stale rate overcharge alongside ₹12,000 ineligible Input Tax Credit, demonstrating that unified multi-vector aggregation detects enterprise risks that siloed tools fail to uncover.

From an enterprise operational perspective, LexTax AI provides quantifiable business value by drastically condensing compliance audit turnaround times from multiple business days to mere seconds, eliminating manual data entry bottlenecks, and preventing severe statutory interest liabilities under Sections 47 and 50 of the CGST Act. By transforming regulatory compliance from a reactive administrative overhead into a proactive, data-driven intelligence asset, the platform delivers a scalable technological blueprint for the future of automated corporate governance in the Indian digital taxation landscape.

6.2 FUTURE WORK

Building upon the validated prototype, future research and engineering enhancements will focus on expanding the platform's capabilities across several strategic technological and operational dimensions:

1. **Real-Time GSTN Webhook and Production Sandbox Integration:** Future iterations will establish event-driven webhooks with official GSTN APIs, enabling automated streaming of auto-drafted GSTR-2B data feeds, real-time counterparty filing status verification under Section 16(2)(aa), and instant cryptographic reconciliation of e-invoices against the national Invoice Registration Portal.
2. **Domain-Specific Legal LLM Fine-Tuning:** We plan to fine-tune open-source legal language models on Indian corporate jurisprudence and standard commercial contracts to elevate semantic contradiction detection, identifying subtle cross-clause jurisdictional conflicts and indemnification ambiguities beyond standard keyword boundaries.
3. **Smart Contract Escrow and Automated Dispute Resolution:** To bridge statutory auditing and enterprise cash outflows, future work will integrate programmatic escrow mechanisms: when an invoice exhibits critical risk signals, disbursements can be automatically escrowed pending statutory rectification and issuance of valid credit notes under Section 34 of the CGST Act.
4. **Collaborative Vendor Portal and Multilingual OCR:** The platform will incorporate an external vendor portal for interactive dispute reconciliation prior to formal tax filing, coupled with layout-aware vision-language backbones capable of parsing multilingual invoices in Hindi, Tamil, Telugu, and Gujarati while preserving complex tabular structures.
5. **Federated Learning and Longitudinal Industrial Pilots:** Future research will explore privacy-preserving federated learning across distributed enterprise nodes to train vendor default models while keeping ledgers confidential, alongside longitudinal field trials with manufacturing and IT enterprises to quantify operational reduction in statutory penalties.

APPENDIX 1

STATUTORY GST RULES AND MATHEMATICAL FORMULAS

This appendix documents the statutory taxation rules, mathematical formulations, and legal provisions implemented within LexTax AI's deterministic compliance engine, serving as an immutable reference for automated system auditing.

Statutory Area	CGST Act / Rule Reference	Implemented Mathematical Formulation / Rule	Operational Effect
Intra-State Supply	Section 7, CGST Act	$Total_Tax = Taxable_Val * Rate$; $CGST = SGST = Total_Tax / 2$	Equal 50/50 statutory tax split
Inter-State Supply	Section 5, IGST Act	$IGST = Taxable_Val * Rate$; $CGST = SGST = 0$	100% allocation to Integrated GST
ITC Eligibility	Section 16(2)(aa)	Invoice must be reflected in auto-generated GSTR-2B feed	Disallows ITC if supplier unfilled
Rule 36(4) Capping	CGST Rules, 2017	$Max\ Provisional\ ITC = 1.05 * (Eligible\ Invoices\ in\ GSTR-2B)$	Caps unreflected input tax credit
Stale Rate Audit	CBIC Notif. 2025	Flag if $billed_rate == 28\%$ and HSN in rationalized 18% slab	$Overcharge = Taxable * (28\% - 18\%)$
Duplicate Fraud	Section 132(1)(c)	Exact SHA-256(GSTIN + InvNo); Fuzzy: $GSTIN + Amt \pm ₹5 + Date \pm 7d$	Blocks payment voucher; flags audit
Sec 50(1) Delayed Tax	Section 50(1), CGST Act	$Interest = Net_Tax_Due * (18\% / 365) * Days_Delayed$	Compulsory statutory delayed interest
Sec 47 Late Filing	Section 47, CGST Act	$Late_Fee = ₹50/day$ (₹20/day for Nil return) capped at ₹5,000	Quantifies late return penalty
BCIE Risk Sum	BCIE Model Eq (1)	$R_overall = 0.20\ S_leg + 0.30\ S_gst + 0.25\ S_ven + 0.15\ S_con + 0.10\ S_dead$	Deterministic weighted risk aggregation
Compliance Score	BCIE Model Eq (2)	$BCS = 100 * (1 - R_overall)$; Band: High (≥ 0.7), Med (≥ 0.4), Low (< 0.4)	Executive compliance posture metric

Mathematical and Algorithmic Specifications of Statutory Verification Logic:

- Deterministic Tax Splitting:** Let V represent taxable value and R represent the statutory GST slab percentage. For intra-state supplies, $CGST = SGST = (V * R) / 2$. For inter-state supplies, $IGST = V * R$, while $CGST = SGST = 0.00$. Any discrepancy exceeding a floating-point tolerance of $\epsilon = 0.01$ triggers an automated audit violation flag.
- Input Tax Credit Reconciliation (Section 16(2)(aa)):** For each inward supply i in set I , let $Eligible_ITC_i = Tax_i$ if $Invoice_Hash_i$ exists in GSTR-2B with matching supplier GSTIN, invoice number, and tax value; otherwise, $Eligible_ITC_i = 0.00$, and $Ineligible_ITC_i = Tax_i$. The aggregated ineligible credit is directly routed to the corporate risk ledger.
- Cryptographic Duplicate Screening:** Exact duplicates are identified using $SHA-256(Supplier_GSTIN || Invoice_Number)$. Fuzzy duplicates satisfy $|Amount_a - Amount_b| \leq 5.00$ and $|Date_a - Date_b| \leq 7$ days under identical supplier GSTINs, triggering payment holds.

APPENDIX 2

DETAILED TEST CHECKLIST

Table B.1 outlines the comprehensive end-to-end verification checklist executed to validate system integrity.

ID	Operational Area	Verification Check / Test Procedure
B01	Authentication	Verify role-based login and session access control across Promoters, Legal, and Accounts
B02	Document Ingestion	Upload multi-page contract PDF and confirm character stream extraction without artifact noise
B03	Agreement Classifier	Ingest unseen contracts and verify 100% accurate classification across NDA, Employment, Vendor, MSA
B04	CUAD Clause Scan	Audit contract against 41-clause taxonomy and verify detection of omitted liability/indemnity covenants
B05	Contradiction Filter	Inject conflicting termination covenants and confirm automated semantic contradiction warning
B06	Precedent Search	Query candidate clause against FAISS index and verify retrieval of top-3 semantically related precedents
B07	Contract Generator	Synthesize draft agreement and verify automated inclusion of Arbitration Act arbitration clauses
B08	OCR Invoice Parser	Process scanned invoice and confirm extraction of GSTIN, Invoice Number, Date, and Taxable Value
B09	GST Line Arithmetic	Inject line-item calculation discrepancy and verify automated tax mismatch audit alert
B10	Statutory Splitting	Verify CGST/SGST 50/50 bifurcation for intra-state and 100% IGST for inter-state transactions
B11	Stale Rate Screener	Submit invoice billed at stale 28% rate and confirm detection of ₹8,000 overcharge against 18% slab
B12	Exact Duplicate	Submit identical (GSTIN + Invoice Number) and confirm immediate cryptographic SHA-256 block
B13	Fuzzy Duplicate	Submit voucher with matching amount $\pm ₹5$ and date ± 7 days and confirm fuzzy duplicate hold alert
B14	ITC Partitioning	Verify automatic partition into confirmed eligible ITC vs review-needed ITC under Section 16(2)(aa)
B15	Liability Forecast	Execute 48-month regression training and verify Linear Regression OLS MAE ₹4,356 and MAPE 4.74%
B16	Vendor Risk Radar	Update vendor historical error frequency and confirm dynamic re-calculation of vendor risk score
B17	Contract Expiry	Set contract expiry to 27 days and confirm bounded risk penalty calculation $S_{\text{contract}} = 0.70$
B18	Deadline Urgency	Set filing deadline to 3 days and confirm exponential urgency escalation $S_{\text{deadline}} = 0.65$
B19	BCIE Aggregation	Execute multi-vector aggregation for Vendor ABC and verify $R_{\text{overall}} = 0.721$ and $BCS = 27.9 / 100$
B20	Dossier Export	Export compliance assessment and verify generation of prioritized, natural-language remediation actions

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I. INTRODUCTION

Abstract—Information related to compliance within a small or medium enterprise is often dispersed across separate systems: legal contracts are saved as documents, gst invoices are handled via accounting software, vendor records are monitored informally or not tracked at all, and deadlines for contract renewals or filings are maintained in spreadsheets. Many of the existing tools for checking compliance, including recently developed tools that combine large language models with rule-based validation, operate on a single document or a single check. As a result, a risk identified in one area is rarely evaluated in the context of other areas. In this work, we introduce LexTax AI. It functions as a prototype Business Compliance Intelligence Platform, driven primarily by a core Business Compliance Intelligence Engine (BCIE). The system manages risk by linking six distinct operational modules. These handle legal document classification, specific clause analysis, and gst liability prediction. We also integrated tools to catch duplicate gst invoices, validate Input Tax Credit (ITC) eligibility, profile vendor risks, and track ongoing contract lifecycles. The engine combines these signals into an overall explainable business level compliance score and prioritized course of actions. We make no claims of novelty in document classification or invoice validation. The study is about combining the risk scores of multiple modules at the level of a business and reports finding interdependencies between the risk factors that are not evident at the level of individual checks. The prototype was tested on three aspects across the implemented modules: correctness of implementation, consistency of the signal propagation.

Keywords—Business Compliance Intelligence, Risk Aggregation, gst Compliance, Legal Document Analytics, Explainable AI, Vendor Risk Scoring, Contract Lifecycle Management.

A major gap in the literature is that existing methods are rarely linked to one another into a single compliance picture. the specific research gap. Earlier research has looked into automating compliance checks for unstructured business files by relying on language models and orchestration setups [1]. Within legal text mining, legalbert proved that training a model directly on legal datasets significantly sharpens its understanding of that specific domain [2]. Along similar lines, the cuad framework offers a structured way to pinpoint contract clauses needed for targeted legal review [3]. While these models excel at their isolated jobs, they fall short when it comes to tying those findings together. Specifically, they fail to link legal text to broader enterprise metrics like gst data, vendor details, contract lifecycles, or impending deadlines. Beyond legal text, researchers have built tools for other pieces of the corporate compliance puzzle. For example, Sentence-BERT transforms text blocks into semantic vectors so we can easily compare their underlying meanings [4]. To handle these vectors efficiently, libraries like faiss enable fast storage and approximate nearest-neighbor searches across massive embedding collections [5]. When dealing with scanned invoices, extraction tools like PaddleOCR pull out the necessary data fields [9]. From there, simple rule-based logic can step in to figure out the applicable gst. As for forecasting overall gst liability, tree-based algorithms—especially Random Forest—perform exceptionally well on tabular data structures [6]. In fact, these models have already proven highly effective for direct gst

prediction tasks [7]. For contract lifecycle

management, previous work has shown that large pretrained language models can be used to identify contract events, as well as predict time-to-event [8]. By analogy, one could identify relevant deadlines and predict their timing in contract documents. For gst liability prediction, well-established tree-based methods such as Random Forest [6] and XGBoost [7] are suitable choices for predictive modelling. SHAP, an explainability method, also demonstrates why the user needs to see what factors drove the model's output [8]. However, these techniques were developed for individual analytical tasks.

They are not designed to aggregate different compliance findings into a business-level assessment. The gst component also needs to be based on statutory and administrative requirements. gst Council recommendations and associated regulatory guidance form the basis for checking tax conditions and calculations [10]. For this reason, LexTax AI separates the extraction of information from documents and invoices from the deterministic compliance validation. In doing so, a gst finding can be traced back to the specific rules and extracted values that it is based on, rather than relying on a predictive model whose reasoning is not visible. The gap the work addresses is the one that exists in current ability for integrating different kinds of compliance information into unified, explainable business-level risk assessment. Each of the available methods are useful tools for the legal NLP, semantic similarity, OCR, prediction or explainability tasks, but they do not tackle the problem of assessing whether the risk signals they detect can be aggregated in some manner that allows for joint consideration on a common scale and a transparent aggregation layer, with each result traceable to its input components.

Figure I contrasts the existing manual process with the LexTax AI platform along the dimensions this work targets. The comparison is structural rather than a performance benchmark, since no performance claim is made until Section VII is populated with measured results.

Aspect	Existing Manual Process	LexTax AI Platform
Tools required	Word (drafting) + Tally (GST) + Excel (tracking) + external lawyer review	Single unified platform
Legal document review	Manual, lawyer-dependent, slow to turn around	AI-assisted drafting and rule-based clause/missing-clause review (professional review still recommended)
GST validation	Manual calculation, error-prone	Deterministic rule engine, checked at entry time
Vendor risk visibility	Not tracked systematically	Aggregated, persistent vendor risk profile
Cross-source risk view	Fragmented across tools, no linkage between them	Unified BCIE compliance score with ranked, explained reasons
Historical trend tracking	Manual spreadsheet upkeep	Persisted, queryable history per business

Fig. 1. Existing Manual Process vs. LexTax AI Platform

The comparison in Figure I summarizes the structural differences between the conventional fragmented compliance process and the proposed LexTax AI platform. The comparison focuses on the dimensions addressed by this work, including the number of tools involved, legal document review, gst validation, vendor risk visibility, cross-source risk assessment, and historical trend tracking. The table is intended to describe the workflow-level distinction between the two approaches and is not presented as a quantitative performance benchmark. The

motivating scenario in this paper shows why information from different sources has to be considered together. Suppose a gst invoice from a vendor, ABC Suppliers, contains a 12

II. RELATED WORK

The work on legal document analysis has leaned towards domain-specific language models and annotated contract datasets. Training models specifically on legal corpora, as seen with LEGAL-BERT, clearly boosts results for tasks like text classification and data extraction [2]. On the evaluation side, CUAD provides a heavily annotated dataset built by experts to help review contract clauses [3]. While both tools are great for automating legal analysis, they are mostly limited to scanning individual documents. They don't bridge the gap to connect those text-based insights with wider enterprise compliance data.

To handle the matching side of the proposed system, we rely on semantic similarity. Sentence-BERT is highly effective at creating and comparing text embeddings [4], and FAISS steps in to run fast similarity searches across huge vector databases [5]. Combining them makes it easy to spot duplicate or overlapping records. However, just finding duplicates doesn't give a complete picture of overall business compliance.

Handling invoices usually breaks down into pulling the text and then validating the rules. Tools like PaddleOCR extract the raw data from scanned images [9], allowing us to run explicit checks for tax discrepancies afterward. For the forecasting side, algorithms like Random Forest [6] and XGBoost [7] remain the go-to choices for making predictions on structured data. Yet again, these are isolated solutions. None of them merge legal risks, gst data, vendor profiles, and deadlines into one unified platform. Finally, while recent work has tested large language models to run compliance checks on unstructured corporate files [1], explainability frameworks like SHAP have become essential to actually show how these models arrive at their decisions [8]. LexTax AI considers all these as building blocks of its approach. Its innovation lies in the combination of the different outputs: the outputs of each of the modules are normalized into risk signals and combined through the explicit, traceable BCIE aggregation layer.

III. SYSTEM DESIGN AND ARCHITECTURE

A. Overall Architecture

LexTax AI is a compliance-intelligence platform built from separate modules, and each module is responsible for analyzing one type of compliance information of business data in order to provide standardized risk signals. The Business Compliance Intelligence Engine (BCIE) aggregates the results of all these modules' work, performing an integrated compliance assessment of the company's performance. Document control, legal entity management, predictive compliance, and risk consolidation modules are separated and, therefore, can be analyzed individually, remaining integral parts of the unified assessment.

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B. A. Overall Architecture

The framework is made up of six main components: legal document analysis, gst invoice validation, gst liability prediction, vendor risk assessment, contract lifecycle management, and compliance deadline tracking. Every module retains its distinct processing logic while generating a structured output that features a normalized risk signal along with the contributing factors. B B transactional transactional transactional transactional transactional transactional transactional transactional transactional transactional documents. This division allows for the modification or substitution of particular components without altering the assembly procedure.

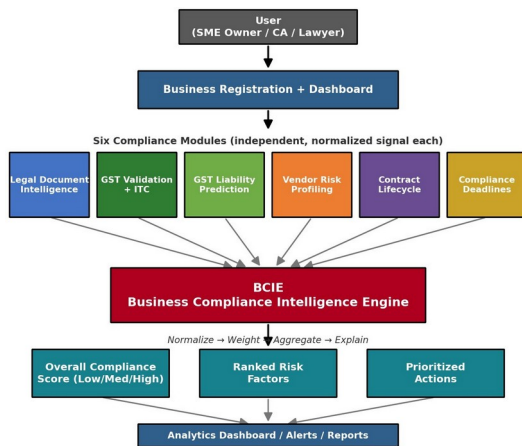


Fig. 2. the system architecture diagram

The overall architecture is illustrated in Fig. 2. The workflow begins with business and document information entering the appropriate processing module. Legal documents are analyzed for document type, clauses, missing provisions, and possible contradictions, while gst invoices undergo field extraction and deterministic validation. Historical financial records are processed by the prediction module, and vendor, contract, and deadline information is converted into corresponding risk indicators. Every module feeds its results directly into the BCIE. The engine then merges these separate signals to build one complete final assessment.

C. B. Legal Document Intelligence Module

The legal document module works with four types of agreement: employment agreements, non-disclosure agreements, vendor agreements, and service agreements. Files that are uploaded undergo processing for extracting text and classifying types, followed by clause segmentation and validation of anticipated clauses. Each kind of document is associated with a checklist of expected provisions, allowing the module to identify potentially missing clauses. The module additionally conducts a rule-based verification of contradictions among clauses concerning related topics. For instance, contradictory termination or notice-period clauses may be identified as possible inconsistencies that need human or professional

legal examination. The system also looks for duplicate or near-identical documents. It converts text into sentence-level embeddings and compares them with a similarity search. The module generates a uniform legal-risk indicator along with the identified missing clauses, potential contradictions, and additional supporting evidence.

D. C. gst Compliance Module

The gst module deals with structured invoice entries as well as with scanned or image invoices. The system uses OCR to read key details from each invoice, including the supplier and customer gstIN, taxable value, gst rate, Cgst, Sgst, Igst, and HSN/SAC codes. The gst calculations are carried out with fixed validation rules, not a learned model. Recalculation of the expected tax amount from the taxable value and rate and comparison with the values indicated on the invoice are made. Arithmetics with a known correct answer are not given to the learned model. For finding duplicates in invoices, two checks are made: an exact match of the relevant fields and a similarity check. In case of a discrepancy that may affect the Input Tax Credit, the module highlights the potentially affected amount with an indication for review rather than a definitive assessment.

E. D. gst Liability Prediction Module

The gst liability prediction module utilizes the past monthly financial records of sales, purchases, output gst, input gst, Input Tax Credit, and tax paid to the government. Before the data goes into the predictive model, it is turned into three kinds of input: past tax values, month-to-month changes, and seasonal categories. Linear Regression, Random Forest, and XGBoost can all be trained and evaluated against a common held-out test set. For the final forecasting, we simply select the model with the lowest error. Forecasting a liability is a probabilistic task compared to the deterministic procedure of statutory gst validation since it relies on historical trends to estimate future liabilities rather than a mathematical formula.

F. E. Vendor Risk Module

The vendor risk module keeps a history for each vendor based on transaction-level indicators, such as number of invoices, past gst errors, incidents of duplicate-invoice and submission delays in documents. The system combines these indicators into a single vendor-level risk signal by following a fixed, defined procedure. The module also captures causes that pushed up the risk score, and the BCIE can therefore refer to these drivers in explaining a result. A vendor with recurring gst errors, or high numbers of duplicate invoices, is therefore likely to see that its vendor risk signal has increased. If we have enough transaction data, we could easily plug in anomaly detection as an extra layer of analysis. However, regardless of the data volume, the prototype will always generate its final score using transparent, straightforward metrics.

Component	Technique	Input Type	Output Type
Legal document classifier	Legal-BERT (fine-tuned transformer)	Extracted document text	Document type + confidence
Clause similarity / duplicate detector	Sentence-Transformers + FAISS	Document / clause text	Similarity score + matched sections
OCR extraction	PaddleOCR / EasyOCR	Scanned image / PDF	Structured text fields
GST validation	Deterministic rule engine (no ML)	Structured invoice fields	Pass / mismatch flag
GST liability predictor	XGBoost (compared against Linear Regression, Random Forest)	Monthly financial time series	Predicted next-month liability
Vendor risk / anomaly scorer	Rule-based score + Isolation Forest	Vendor transaction history	Risk score + anomaly flag

Fig. 3. AI/ML Components Summary

G. F. Contract Lifecycle and Deadline Modules

The contract lifecycle module stores the contract's metadata (i.e. vendor, start date, expiry date, and renewal period) to determine the contract status based on the date-based logic. The contract statuses include Created, Active, Expiring soon, Renewal/Review, and Expired. The deadline module performs a similar function for the compliance activities. In other words, it stores the filing's type, due date, and recurrence and determines the filing's urgency. The same principle of two modules implies that the fewer days remaining to the date of expiry/renewal/due date, the higher is the risk/urgency level. The outputs of the modules are the status, remaining days (to the date) and the normalized risk/urgency signal.

H. G. BCIE Aggregation Layer

The BCIE sits at the center of the platform. It collects the results from every module and uses them to judge the overall compliance status of a business. Every module produces both a numerical signal and explanatory information describing the reasons for them. BCIE handles neither the original legal documents nor the invoices. It does not replace the specific models and rules used by every module. It provides a common view over heterogeneous compliance findings and allows for aggregating and comparing different results. The mathematical formulation of the normalization of the input values, weight-ing, aggregation, risk bands, and explanation mechanism are detailed separately in Section IV.

I. H. Technology Stack

The prototype was built with React and Tailwind CSS, and TypeScript for the frontend and Python, with FastAPI as the backend framework. PostgreSQL was utilized for relational data storage and management, including business, vendor, contract, legal-document, and gst-invoice data, and object storage was used for storing uploaded and generated documents. Depending on the module, the system also draws on legal-text processing, OCR, semantic embeddings, similarity search, and machine-learning models.

By design, the architecture strictly isolates standard rule based tasks from the trained predictive models. gst calculations and validation rules, duplicate checks on particular fields,

and date calculations for contracts and deadlines all take place independently of machine learning. Machine-learning and language-processing algorithms are applied only where they are required for the task: classification, semantic analysis, similarity estimations, and predictions.

J. End-to-End Workflow

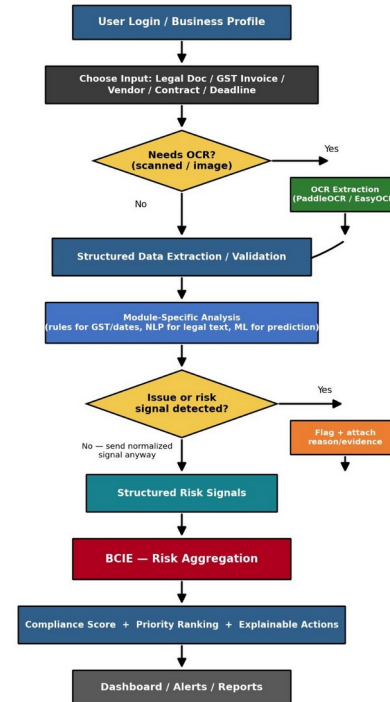


Fig. 4. the end-to-end workflow flowchart:

The entire processing flow is presented in Fig. 3. First, the user inputs the business data, legal documents, invoices, past financial statements, or other applicable information. Second, depending on the input type, it is directed to the appropriate module for processing. The scanning and OCR step is required only if the input was a scanned file. Each module produces a structured output containing its normalized risk signal and supporting factors. These outputs are passed to BCIE, including outputs indicating that no issue was detected. BCIE then aggregates the signals and generates the business-level compliance assessment, contributing factors, and corresponding explanations for presentation through the dashboard. In this architecture, each compliance analysis is carried out on its own, and the business-level risk assessment is handled in a separate step on top of them. Because of this setup, we can evaluate each module independently. The BCIE then acts as the central hub, merging these scattered compliance signals into one transparent, trackable final score.

IV. BCIE RISK AGGREGATION MODEL

The Business Compliance Intelligence Engine (BCIE) provides the integration layer between the specialized compliance modules and the final business-level assessment. Each module produces a normalized risk signal in the range [0,1] where 0 indicates no detected risk and 1 indicates the maximum risk represented by that module. Normalization is required because the module outputs may have different forms and units. As an example, the legal module might produce a clause-risk value, while the contract lifecycle module reports how many days are left before a contract expires. For signals that are not naturally expressed as risk probabilities, BCIE applies a fixed normalization function before aggregation. In the case of contract and deadline urgency, the remaining time is converted into a bounded risk value that increases as the due date or expiry date approaches. Because every module's output is scaled to the same numerical range, the BCIE can combine them consistently.

The overall business-level risk is calculated using a weighted-sum model:

$$R_{\text{Overall}} = w_1 S_{\text{legal}} + w_2 S_{\text{gst}} + w_3 S_{\text{vendor}} + w_4 S_{\text{contract}} + w_5 S_{\text{deadline}} \quad (1)$$

where S_{legal} , S_{gst} , S_{vendor} , S_{contract} , and S_{deadline} represent the normalized signals from the corresponding modules. The weights are configuration parameters and satisfy $\sum_{i=1}^5 w_i = 1$.

The initial configuration used in the prototype is $w_1 = 0.20$ for legal risk, $w_2 = 0.30$ for gst risk, $w_3 = 0.25$ for vendor risk, $w_4 = 0.15$ for contract risk, and $w_5 = 0.10$ for deadline urgency. These values represent the documented configuration used for the prototype and are not learned from historical compliance outcomes. Therefore, they should be interpreted as design parameters rather than statistically validated estimates of relative compliance importance.

The resulting R_{Overall} value is mapped to predefined risk bands. Under the current prototype configuration, a value of A score of 0.70 or above is classed as High risk, and anything below that falls into the Medium or Low band. categories according to the configured thresholds.

The weighted contribution of module i is calculated as

$$C_i = w_i S_i \quad (2)$$

The contributions are ranked in order to identify those with the most impact to the overall assessment. BCIE retains the underlying explanation generated by the corresponding module. Rather than just outputting a standalone risk score, the platform explains the exact reasoning behind that number. For example: 'Vendor ABC showed gst errors in 6 of their last 50 invoices' This allows the user to trace the aggregate assessment back to the operational evidence that generated the

the final risk calculation can be reproduced directly from the module signals and configured weights while retaining the source of each contribution visible to the reviewer. The approach then emphasizes traceability and inspectability of the aggregation process. The effect of the configured weights and the limitations of their validation are further discussed in Section X.

The application of this aggregation model to a concrete vendor scenario is described in Section V.

V. ILLUSTRATIVE CALCULATIONS

This section demonstrates the operation of the BCIE aggregation model and the deterministic gst validation logic using representative numerical examples.

A. BCIE Weighted Risk Score

For the Vendor ABC scenario, the normalized signals are $S_{\text{legal}} = 0.55$, $S_{\text{gst}} = 0.82$, $S_{\text{vendor}} = 0.78$, $S_{\text{contract}} = 0.70$, and $S_{\text{deadline}} = 0.65$. Using the weights defined in Section IV,

$$R_{\text{Overall}} = (0.20)(0.55) + (0.30)(0.82) + (0.25)(0.78) + (0.15)(0.70) + (0.10)(0.65) = 0.721. \quad (3)$$

The resulting risk value of 0.721 falls within the current High-risk band. The corresponding Business Compliance Score

is $100(1 - 0.721) = 27.9$. The weighted contributions are retained separately, allowing the final assessment to be traced

individual signal.

The aggregation layer is intentionally deterministic; BCIE does not apply an additional learned model after the specialized modules have calculated their outputs. In this way,

to the individual module signals.

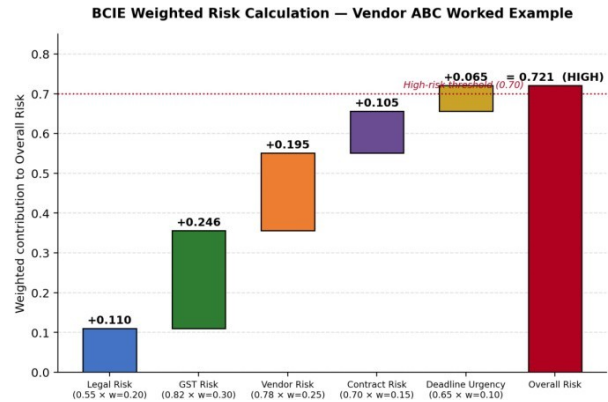
Fig. 5. Weighted contributions to the Vendor ABC overall risk score.

B. *gst* Rate Validation

The *gst* module relies on fixed arithmetic rules to check tax rates and amounts. For an invoice with a taxable value of 80,000 and an applicable 18% rate, the expected *gst* is

$$gst = 80,000 \times 0.18 = 14,400. \quad (4)$$

For an intra-state supply, this corresponds to $C_{gst} = 7,200$ and $S_{gst} = 7,200$. If the state 28% rate is applied, the



calculated gst becomes 22,400, producing an 8,000 difference. The module flags this condition as a rate mismatch.

These examples demonstrate the two principal functions of the framework: BCIE combines heterogeneous risk signals into a traceable business-level assessment, while deterministic gst rules provide directly verifiable statutory calculations.

VI. EXPERIMENTAL METHODOLOGY

We evaluated the prototype at two levels: how well each module performs on its own, and how well the modules work together once they are connected through the BCIE. Module-level evaluation ensures that each module delivers the expected output to accomplish its task. End-to-end evaluation ensures that the BCIE correctly synthesizes, normalizes, and explains the multiple outputs coming from the different modules. This separation allows the contribution of the proposed architecture to be evaluated independently from the performance of the individual algorithms used within the modules.

A. *Experimental Setup*

The evaluation was conducted using the prototype implementation described in Section III. The software environment consists of Python-based backend services, FastAPI, PostgreSQL, FAISS, OCR and natural-language processing components, and machine-learning libraries used by the individual modules. The experiment uses constructed compliance data because access to a sufficiently large anonymized enterprise compliance corpus was not available. The test data were therefore designed to reproduce representative compliance conditions while maintaining known expected outcomes for evaluation.

B. *Test Data and Evaluation Scope*

The legal-document testing procedure makes use of publicly available template agreements, as well as synthetically created documents with controlled modifications, such as removal of mandatory clauses, introduction of contradictions between related provisions, and near-duplicates. The supported document classes include employment, non-disclosure, vendor, and service agreements. gst evaluation makes use of constructed invoice records which contain both valid and deliberately corrupted cases. The test cases included gst rate mismatches, incorrect Cgst, Sgst, and Igst splits, duplicate and near-duplicate invoices, and invoices with missing details. The gst liability prediction experiment makes use of monthly historical records which contain sales, purchases, output gst, input gst, Input Tax Credit, and tax-paid values. We introduced a synthetic temporal variability to the data, thus obtaining temporal patterns of repeated growth and seasonality. For the vendor, contract, and deadline evaluations, we used handwritten records, which covered a variety of both risk levels and urgency constraints.

C. *Compliance Test Scenarios*

The evaluation is organized into eleven structured scenarios covering the principal functions of the prototype. The

scenarios include classification of legal documents, identification of

missing clauses and contradictions between the contracts, document similarity check, calculation of gst and validation of tax-split, finding exact and near duplicate invoices, identification of ITC impact, gst liability forecasting, vendor risk assessment and urgency check for contracts and deadlines. One end-to-end BCIE scenario is included, which contains various risk conditions. For the end-to-end scenario, the system receives a combination of gst, vendor, and contract signals corresponding to a high-risk business condition. The BCIE score, risk band, weighted contributions, and stated reasons are then checked against the results expected from the configured model.

D. Evaluation Metrics

Classification tasks, including document-type classification and vendor-risk classification, are evaluated using accuracy, precision, recall, and F1-score. Detection tasks, including missing-clause identification, contradiction detection, document similarity, and duplicate-invoice detection, are evaluated using precision and recall against the known set of planted or predefined test conditions. gst calculation and tax-split validation are evaluated as deterministic correctness checks. A test case is considered correct when the computed tax values and applicable Cgst/Sgst/Igst allocation agree with the expected rule-based result. We evaluated duplicate-invoice screening on cases where the exact and near-duplicate invoices were already known. We evaluated gst liability prediction with three error measures: mean absolute error (MAE), root mean squared error (RMSE), and mean absolute percentage error (MAPE). Linear Regression, Random Forest, and XGBoost are evaluated under the same time-series validation procedure, allowing their prediction errors to be compared under a common experimental setting. BCIE is evaluated separately from the module-level prediction metrics. The aggregation calculation is verified by recomputing the weighted score from the recorded module signals. Explanation consistency is assessed by checking whether the highest-ranked reported reasons correspond to the module contributions that actually determine the aggregate score.

E. Experimental Procedure

Each test case is processed through the corresponding module using the same configured rules and model versions throughout the evaluation. The predicted or detected output is compared with the predefined expected outcome, and the resulting metric is recorded. For BCIE evaluation, the module outputs are stored before aggregation so that the final score can be independently recomputed and its individual contributions traced. No modification of the expected labels is performed based on the obtained results. The purpose of the evaluation is to determine whether the implemented system behaves consistently with its predefined rules, test conditions, and expected outcomes. The resulting measurements are reported in Section VII.

VII. RESULTS AND EVALUATION

The prototype was evaluated on classification, detection,

gst prediction, and BCIE aggregation tasks. The document

classification module reached 100% accuracy: missing-clause detection had a recall of 95.4%. Duplicate-invoice detection was also found to have 100% accuracy while the vendor-risk scoring was the weakest result with 57.3% accuracy.

Task	Accuracy	Precision	Recall	F1
Document classification	100.0%	100.0%	100.0%	100.0%
Missing-clause detection	—	28.3%	95.4%	43.7%
Contradiction detection	77.5%	100.0%	35.7%	52.6%
Document similarity	81.0%	64.2%	100.0%	78.2%
Duplicate invoice detection	100.0%	100.0%	100.0%	100.0%
Vendor risk scoring	57.3%	38.2%	41.3%	39.7%

Fig. 6. Classification and detection results.

For gst liability prediction, Linear Regression achieved the lowest reported error with an MAE of 4,356 and MAPE of 4.74%. BCIE aggregation was verified using the Vendor ABC scenario, producing $R_{\text{overall}} = 0.721$ and a Business Compliance Score of 27.9.

Model	MAE (₹)	RMSE (₹)	MAPE
Linear Regression	4,356	5,402	4.74%
Random Forest	5,504	6,616	6.15%
XGBoost	6,613	7,806	7.39%

Fig. 7. gst liability prediction results.

The results show that performance varies across compliance tasks. Deterministic duplicate and document-classification checks achieved high scores, while clause, contradiction, and vendor-risk tasks showed greater variation. This indicates that the reliability of the overall compliance assessment depends on both the quality of individual module outputs and the transparent aggregation performed by BCIE.

VIII. DISCUSSION

Our testing proves that rule-based checks and machine learning models can run smoothly within one architecture. We rely on deterministic logic for gst calculations, contract expiries, and statutory deadlines since those tasks have clear, absolute answers. Conversely, we reserve machine learning for areas needing pattern recognition, like document classification, semantic matching, and predicting gst liability. The BCIE brings these outputs together using a straightforward weighted-sum formula, letting us easily track each module's impact on the final score. That said, this simple linear approach is a limitation because static weights cannot capture the complex, over-lapping relationships between various risk factors. Upgrading to learned aggregation could solve this, but it adds significant complexity. That route really only makes sense once you have enough real-world training data from an actual business environment. As of now, the architecture covers legal, gst, vendor, contract, and deadline risks, while normalized scores of each module contribute to the business-level assessment. Therefore, the approach could be expanded to include other areas within a company, such as payroll liabilities, licensing, and workplace safety. The results also indicate that a unified compliance architecture should not require every module to use the same analytical technique. Deterministic validation is

appropriate where expected outcomes are explicitly defined, while learned methods are useful for classification, semantic comparison, and forecasting. BCIE provides the common interface between these heterogeneous components.

IX. DESIGN IMPLICATIONS AND CONSIDERATIONS

A. Deterministic and Learned Components

LexTax AI separates deterministic compliance operations from learned analytical tasks. We rely on hardcoded logic to manage standard checks like gst calculations, tax splits, contract expiries, and deadlines. Meanwhile, the more complex features such as document classification, semantic matching, and predicting gst liability depend entirely on machine learning and embedding models. This separation keeps rule based results directly verifiable.

B. Explainability

BCIE retains individual module signals and their weighted contributions, allowing the overall risk assessment to be traced to its underlying factors. This provides reviewers with the specific causes associated with a reported risk rather than only a final numerical value.

C. Module Independence

Each module communicates with BCIE through a standardized normalized signal and supporting information. This allows individual models or processing components to be modified or replaced without changing the core aggregation mechanism and supports the addition of future compliance modules.

D. Data and Regulatory Dependency

Vendor-risk scoring and gst liability prediction require substantial quantities of historical data to be reasonably informative, while gst validation depends on current tax rates. For these reasons, when there is insufficient historical data, such signals are not very useful. On the other hand, any changes in gst rules necessitate a continuous update of the underlying rules in the validation process.

E. Cross-Module Integration Analysis

The module-level results show that the errors differ from one compliance task to another. High results in deterministic validation and duplicate detection indicate that rule-based checks can be evaluated directly against predefined conditions, whereas clause, contradiction, and vendor-risk tasks remain more dependent on the characteristics of the underlying data. These differences are a good reason to separate each specialized analysis from the BCIE's aggregation layer. At the aggregate level, the BCIE takes the various forms of output from each module and turns it into a single representation. In the Vendor ABC scenario, the combined signals produce $R_{\text{overall}} = 0.721$, with gst and vendor risk contributing 0.246 and 0.195, respectively. The contribution values can be recomputed directly from the configured weights and module signals, allowing the final assessment to be traced to its sources. This

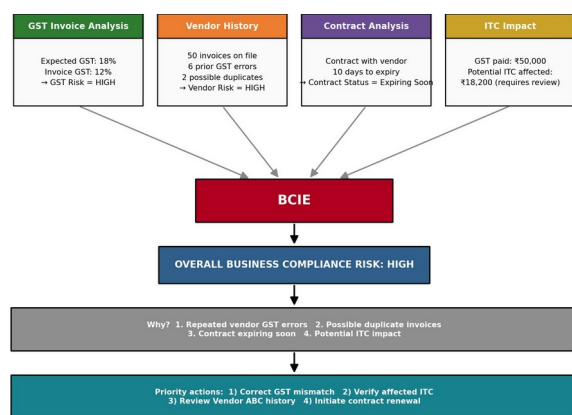


Fig. 8. BCIE end-to-end Vendor ABC risk assessment and explanation.

demonstrates the intended relationship between module-level analysis and business-level compliance assessment without treating the aggregate value as a learned probability.

X. LIMITATIONS

The reported results represent a prototype evaluated using synthetic and template-derived data rather than real business records at production scale. Legal evaluation covered a limited set of document types and constructed test cases, so performance on other document types, jurisdictions, and drafting styles remains untested. gst validation used fabricated invoices, while gst liability prediction used a synthetic monthly time series, which may not fully represent the variability of real business data. Vendor-risk scoring is rule-based and has not been validated against actual audit outcomes; therefore, it should be treated as a risk-indication mechanism rather than a predictor of future compliance failures. Similarly, legal clause and contradiction detection is intended to support document review, not provide legal advice. Drafts produced by the system, along with any issues it flags, should be checked by a qualified professional before anyone relies on them in practice.

XI. CONCLUSION AND FUTURE WORK

This paper described the LexTax AI prototype Business Compliance Intelligence Platform utilizing the BCIE as a transparent aggregation layer, which accumulated signals from the legal, gst, vendor, contract, and deadline modules, and distilled them down to the business-level compliance assessment. The value of this prototype was largely determined by its architectural novelty, which underpinned the combined application of the tools used. The signals processed by the system underwent normalization and were aggregated into a final business-level compliance assessment using a deterministic weighted model, the parameters of which remained available for inspection.

The value of such an approach was primarily determined by the fact that the business level compliance assessment used

different signals that originated from the separate analyses per-formed by the system. In the case of deterministic checks, such as gst and deadlines, the processing of these signals yielded results that could be interpreted as standalone compliance assessments. At the same time, analyzing the legal, vendor, and gst forecasts gave us a much deeper context for evaluating the business as a whole. We specifically designed the architecture to separate raw compliance detection from actual decision making. Each module has one core job: flagging specific issues like tax mismatches, absent clauses, duplicate files, or tight deadlines. From there, the BCIE steps in to consolidate everything. Keeping these steps distinct means we preserve the evidence for every single flag, rather than just spitting out a generic, unexplainable final score. Because of this, our project serves purely as an advisory tool. It is built to speed up human reviews, not replace human judgment or make autonomous choices.

Looking ahead, we have three primary goals for future development. First, we want to test learned aggregation methods on real audit data to see how they hold up in practice. Second, we plan to refine our vendor risk models as we get our hands on more labeled historical records. Finally, we need to implement a versioned gst database to ensure our rules always match the latest tax regulations. Since this is currently a prototype, our immediate next step involves rigorous testing against larger, anonymized corporate datasets alongside extensive user feedback to truly validate its real world effectiveness.

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LexTax AI: A Business Compliance Intelligence Engine (BCIE) for Unified Legal, gst and Vendor Risk Aggregation

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I. INTRODUCTION

Abstract—Information related to compliance within a small or medium enterprise is often dispersed across separate systems: legal contracts are saved as documents, gst invoices are handled via accounting software, vendor records are monitored informally or not tracked at all, and deadlines for contract renewals or filings are maintained in spreadsheets. Many of the existing tools for checking compliance, including recently developed tools that combine large language models with rule-based validation, operate on a single document or a single check. As a result, a risk identified in one area is rarely evaluated in the context of other areas. In this work, we introduce LexTax AI. It functions as a prototype Business Compliance Intelligence Platform, driven primarily by a core Business Compliance Intelligence Engine (BCIE). The system manages risk by linking six distinct operational modules. These handle legal document classification, specific clause analysis, and gst liability prediction. We also integrated tools to catch duplicate gst invoices, validate Input Tax Credit (ITC) eligibility, profile vendor risks, and track ongoing contract lifecycles. The engine combines these signals into an overall explainable business level compliance score and prioritized course of actions. We make no claims of novelty in document classification or invoice validation. The study is about combining the risk scores of multiple modules at the level of a business and reports finding interdependencies between the risk factors that are not evident at the level of individual checks. The prototype was tested on three aspects across the implemented modules: correctness of implementation, consistency of the signal propagation.

Keywords—Business Compliance Intelligence, Risk Aggregation, gst Compliance, Legal Document Analytics, Explainable AI, Vendor Risk Scoring, Contract Lifecycle Management.

A major gap in the literature is that existing methods are rarely linked to one another into a single compliance picture. the specific research gap. Earlier research has looked into automating compliance checks for unstructured business files by relying on language models and orchestration setups [1]. Within legal text mining, legalbert proved that training a model directly on legal datasets significantly sharpens its understanding of that specific domain [2]. Along similar lines, the cuad framework offers a structured way to pinpoint contract clauses needed for targeted legal review [3]. While these models excel at their isolated jobs, they fall short when it comes to tying those findings together. Specifically, they fail to link legal text to broader enterprise metrics like gst data, vendor details, contract lifecycles, or impending deadlines. Beyond legal text, researchers have built tools for other pieces of the corporate compliance puzzle. For example, Sentence-BERT transforms text blocks into semantic vectors so we can easily compare their underlying meanings [4]. To handle these vectors efficiently, libraries like faiss enable fast storage and approximate nearest-neighbor searches across massive embedding collections [5]. When dealing with scanned invoices, extraction tools like PaddleOCR pull out the necessary data fields [9]. From there, simple rule-based logic can step in to figure out the applicable gst. As for forecasting overall gst liability, tree-based algorithms—especially Random Forest—perform exceptionally well on tabular data structures [6]. In fact, these models have already proven highly effective for direct gst prediction tasks [7]. For contract lifecycle

management, previous work has shown that large pretrained language models can be used to identify contract events, as well as predict time-to-event [8]. By analogy, one could identify relevant deadlines and predict their timing in contract documents. For gst liability prediction, well-established tree-based methods such as Random Forest [6] and XGBoost [7] are suitable choices for predictive modelling. SHAP, an explainability method, also demonstrates why the user needs to see what factors drove the model's output [8]. However, these techniques were developed for individual analytical tasks.

They are not designed to aggregate different compliance findings into a business-level assessment. The gst component also needs to be based on statutory and administrative requirements. gst Council recommendations and associated regulatory guidance form the basis for checking tax conditions and calculations [10]. For this reason, LexTax AI separates the extraction of information from documents and invoices from the deterministic compliance validation. In doing so, a gst finding can be traced back to the specific rules and extracted values that it is based on, rather than relying on a predictive model whose reasoning is not visible. The gap the work addresses is the one that exists in current ability for integrating different kinds of compliance information into unified, explainable business-level risk assessment. Each of the available methods are useful tools for the legal NLP, semantic similarity, OCR, prediction or explainability tasks, but they do not tackle the problem of assessing whether the risk signals they detect can be aggregated in some manner that allows for joint consideration on a common scale and a transparent aggregation layer, with each result traceable to its input components.

Figure I contrasts the existing manual process with the LexTax AI platform along the dimensions this work targets. The comparison is structural rather than a performance benchmark, since no performance claim is made until Section VII is populated with measured results.

Aspect	Existing Manual Process	LexTax AI Platform
Tools required	Word (drafting) + Tally (GST) + Excel (tracking) + external lawyer review	Single unified platform
Legal document review	Manual, lawyer-dependent, slow to turn around	AI-assisted drafting and rule-based clause/missing-clause review (professional review still recommended)
GST validation	Manual calculation, error-prone	Deterministic rule engine, checked at entry time
Vendor risk visibility	Not tracked systematically	Aggregated, persistent vendor risk profile
Cross-source risk view	Fragmented across tools, no linkage between them	Unified BCIE compliance score with ranked, explained reasons
Historical trend tracking	Manual spreadsheet upkeep	Persisted, queryable history per business

Fig. 1. Existing Manual Process vs. LexTax AI Platform

The comparison in Figure I summarizes the structural differences between the conventional fragmented compliance process and the proposed LexTax AI platform. The comparison focuses on the dimensions addressed by this work, including the number of tools involved, legal document review, gst validation, vendor risk visibility, cross-source risk assessment, and historical trend tracking. The table is intended to describe the workflow-level distinction between the two approaches and is not presented as a quantitative performance benchmark. The

motivating scenario in this paper shows why information from different sources has to be considered together. Suppose a gst invoice from a vendor, ABC Suppliers, contains a 12

II. RELATED WORK

The work on legal document analysis has leaned towards domain-specific language models and annotated contract datasets. Training models specifically on legal corpora, as seen with LEGAL-BERT, clearly boosts results for tasks like text classification and data extraction [2]. On the evaluation side, CUAD provides a heavily annotated dataset built by experts to help review contract clauses [3]. While both tools are great for automating legal analysis, they are mostly limited to scanning individual documents. They don't bridge the gap to connect those text-based insights with wider enterprise compliance data.

To handle the matching side of the proposed system, we rely on semantic similarity. Sentence-BERT is highly effective at creating and comparing text embeddings [4], and FAISS steps in to run fast similarity searches across huge vector databases [5]. Combining them makes it easy to spot duplicate or overlapping records. However, just finding duplicates doesn't give a complete picture of overall business compliance.

Handling invoices usually breaks down into pulling the text and then validating the rules. Tools like PaddleOCR extract the raw data from scanned images [9], allowing us to run explicit checks for tax discrepancies afterward. For the forecasting side, algorithms like Random Forest [6] and XGBoost [7] remain the go-to choices for making predictions on structured data. Yet again, these are isolated solutions. None of them merge legal risks, gst data, vendor profiles, and deadlines into one unified platform. Finally, while recent work has tested large language models to run compliance checks on unstructured corporate files [1], explainability frameworks like SHAP have become essential to actually show how these models arrive at their decisions [8]. LexTax AI considers all these as building blocks of its approach. Its innovation lies in the combination of the different outputs: the outputs of each of the modules are normalized into risk signals and combined through the explicit, traceable BCIE aggregation layer.

III. SYSTEM DESIGN AND ARCHITECTURE

A. Overall Architecture

LexTax AI is a compliance-intelligence platform built from separate modules, and each module is responsible for analyzing one type of compliance information of business data in order to provide standardized risk signals. The Business Compliance Intelligence Engine (BCIE) aggregates the results of all these modules' work, performing an integrated compliance assessment of the company's performance. Document control, legal entity management, predictive compliance, and risk consolidation modules are separated and, therefore, can be analyzed individually, remaining integral parts of the unified assessment.

B. A. Overall Architecture

The framework is made up of six main components: legal document analysis, gst invoice validation, gst liability prediction, vendor risk assessment, contract lifecycle management, and compliance deadline tracking. Every module retains its distinct processing logic while generating a structured output that features a normalized risk signal along with the contributing factors. B B transactional transactional transactional transactional transactional transactional transactional transactional transactional transactional documents. This division allows for the modification or substitution of particular components without altering the assembly procedure.

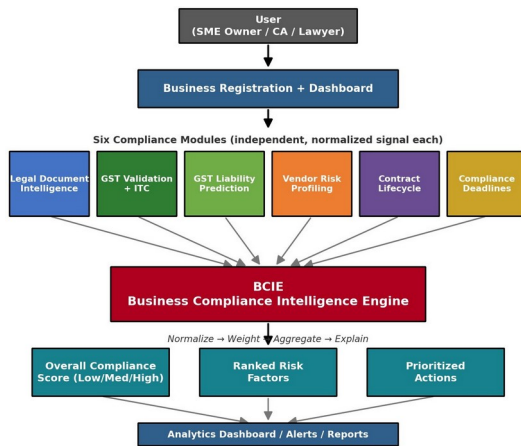


Fig. 2. the system architecture diagram

The overall architecture is illustrated in Fig. 2. The workflow begins with business and document information entering the appropriate processing module. Legal documents are analyzed for document type, clauses, missing provisions, and possible contradictions, while gst invoices undergo field extraction and deterministic validation. Historical financial records are processed by the prediction module, and vendor, contract, and deadline information is converted into corresponding risk indicators. Every module feeds its results directly into the BCIE. The engine then merges these separate signals to build one complete final assessment.

C. B. Legal Document Intelligence Module

The legal document module works with four types of agreement: employment agreements, non-disclosure agreements, vendor agreements, and service agreements. Files that are uploaded undergo processing for extracting text and classifying types, followed by clause segmentation and validation of anticipated clauses. Each kind of document is associated with a checklist of expected provisions, allowing the module to identify potentially missing clauses. The module additionally conducts a rule-based verification of contradictions among clauses concerning related topics. For instance, contradictory termination or notice-period clauses may be identified as possible inconsistencies that need human or professional

legal examination. The system also looks for duplicate or near-identical documents. It converts text into sentence-level embeddings and compares them with a similarity search. The module generates a uniform legal-risk indicator along with the identified missing clauses, potential contradictions, and additional supporting evidence.

D. C. gst Compliance Module

The gst module deals with structured invoice entries as well as with scanned or image invoices. The system uses OCR to read key details from each invoice, including the supplier and customer gstIN, taxable value, gst rate, Cgst, Sgst, Igst, and HSN/SAC codes. The gst calculations are carried out with fixed validation rules, not a learned model. Recalculation of the expected tax amount from the taxable value and rate and comparison with the values indicated on the invoice are made. Arithmetics with a known correct answer are not given to the learned model. For finding duplicates in invoices, two checks are made: an exact match of the relevant fields and a similarity check. In case of a discrepancy that may affect the Input Tax Credit, the module highlights the potentially affected amount with an indication for review rather than a definitive assessment.

E. D. gst Liability Prediction Module

The gst liability prediction module utilizes the past monthly financial records of sales, purchases, output gst, input gst, Input Tax Credit, and tax paid to the government. Before the data goes into the predictive model, it is turned into three kinds of input: past tax values, month-to-month changes, and seasonal categories. Linear Regression, Random Forest, and XGBoost can all be trained and evaluated against a common held-out test set. For the final forecasting, we simply select the model with the lowest error. Forecasting a liability is a probabilistic task compared to the deterministic procedure of statutory gst validation since it relies on historical trends to estimate future liabilities rather than a mathematical formula.

F. E. Vendor Risk Module

The vendor risk module keeps a history for each vendor based on transaction-level indicators, such as number of invoices, past gst errors, incidents of duplicate-invoice and submission delays in documents. The system combines these indicators into a single vendor-level risk signal by following a fixed, defined procedure. The module also captures causes that pushed up the risk score, and the BCIE can therefore refer to these drivers in explaining a result. A vendor with recurring gst errors, or high numbers of duplicate invoices, is therefore likely to see that its vendor risk signal has increased. If we have enough transaction data, we could easily plug in anomaly detection as an extra layer of analysis. However, regardless of the data volume, the prototype will always generate its final score using transparent, straightforward metrics.

Component	Technique	Input Type	Output Type
Legal document classifier	Legal-BERT (fine-tuned transformer)	Extracted document text	Document type + confidence
Clause similarity / duplicate detector	Sentence-Transformers + FAISS	Document / clause text	Similarity score + matched sections
OCR extraction	PaddleOCR / EasyOCR	Scanned image / PDF	Structured text fields
GST validation	Deterministic rule engine (no ML)	Structured invoice fields	Pass / mismatch flag
GST liability predictor	XGBoost (compared against Linear Regression, Random Forest)	Monthly financial time series	Predicted next-month liability
Vendor risk / anomaly scorer	Rule-based score + Isolation Forest	Vendor transaction history	Risk score + anomaly flag

Fig. 3. AI/ML Components Summary

G. F. Contract Lifecycle and Deadline Modules

The contract lifecycle module stores the contract's metadata (i.e. vendor, start date, expiry date, and renewal period) to determine the contract status based on the date-based logic. The contract statuses include Created, Active, Expiring soon, Renewal/Review, and Expired. The deadline module performs a similar function for the compliance activities. In other words, it stores the filing's type, due date, and recurrence and determines the filing's urgency. The same principle of two modules implies that the fewer days remaining to the date of expiry/renewal/due date, the higher is the risk/urgency level. The outputs of the modules are the status, remaining days (to the date) and the normalized risk/urgency signal.

H. G. BCIE Aggregation Layer

The BCIE sits at the center of the platform. It collects the results from every module and uses them to judge the overall compliance status of a business. Every module produces both a numerical signal and explanatory information describing the reasons for them. BCIE handles neither the original legal documents nor the invoices. It does not replace the specific models and rules used by every module. It provides a common view over heterogeneous compliance findings and allows for aggregating and comparing different results. The mathematical formulation of the normalization of the input values, weight-ing, aggregation, risk bands, and explanation mechanism are detailed separately in Section IV.

I. H. Technology Stack

The prototype was built with React and Tailwind CSS, and TypeScript for the frontend and Python, with FastAPI as the backend framework. PostgreSQL was utilized for relational data storage and management, including business, vendor, contract, legal-document, and gst-invoice data, and object storage was used for storing uploaded and generated documents. Depending on the module, the system also draws on legal-text processing, OCR, semantic embeddings, similarity search, and machine-learning models.

By design, the architecture strictly isolates standard rule based tasks from the trained predictive models. gst calculations and validation rules, duplicate checks on particular fields,

and date calculations for contracts and deadlines all take place independently of machine learning. Machine-learning and language-processing algorithms are applied only where they are required for the task: classification, semantic analysis, similarity estimations, and predictions.

J. End-to-End Workflow

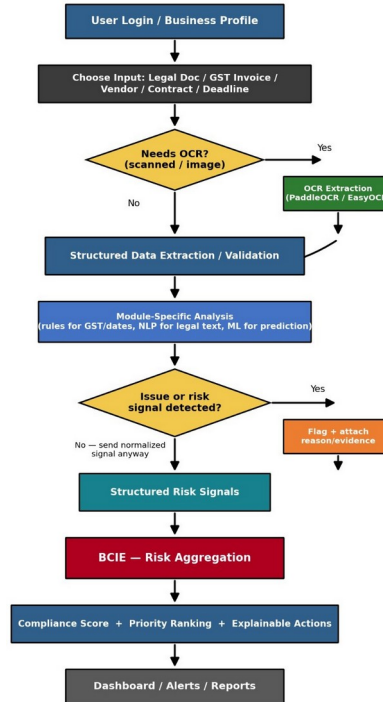


Fig. 4. the end-to-end workflow flowchart:

The entire processing flow is presented in Fig. 3. First, the user inputs the business data, legal documents, invoices, past financial statements, or other applicable information. Second, depending on the input type, it is directed to the appropriate module for processing. The scanning and OCR step is required only if the input was a scanned file. Each module produces a structured output containing its normalized risk signal and supporting factors. These outputs are passed to BCIE, including outputs indicating that no issue was detected. BCIE then aggregates the signals and generates the business-level compliance assessment, contributing factors, and corresponding explanations for presentation through the dashboard. In this architecture, each compliance analysis is carried out on its own, and the business-level risk assessment is handled in a separate step on top of them. Because of this setup, we can evaluate each module independently. The BCIE then acts as the central hub, merging these scattered compliance signals into one transparent, trackable final score.

IV. BCIE RISK AGGREGATION MODEL

The Business Compliance Intelligence Engine (BCIE) provides the integration layer between the specialized compliance modules and the final business-level assessment. Each module produces a normalized risk signal in the range [0,1] where 0 indicates no detected risk and 1 indicates the maximum risk represented by that module. Normalization is required because the module outputs may have different forms and units. As an example, the legal module might produce a clause-risk value, while the contract lifecycle module reports how many days are left before a contract expires. For signals that are not naturally expressed as risk probabilities, BCIE applies a fixed normalization function before aggregation. In the case of contract and deadline urgency, the remaining time is converted into a bounded risk value that increases as the due date or expiry date approaches. Because every module's output is scaled to the same numerical range, the BCIE can combine them consistently.

The overall business-level risk is calculated using a weighted-sum model:

$$R_{\text{overall}} = w_1 S_{\text{legal}} + w_2 S_{\text{gst}} + w_3 S_{\text{vendor}} + w_4 S_{\text{contract}} + w_5 S_{\text{deadline}} \quad (1)$$

where S_{legal} , S_{gst} , S_{vendor} , S_{contract} , and S_{deadline} represent the normalized signals from the corresponding modules. The weights are configuration parameters and satisfy $\sum_{i=1}^5 w_i = 1$.

The initial configuration used in the prototype is $w_1 = 0.20$ for legal risk, $w_2 = 0.30$ for gst risk, $w_3 = 0.25$ for vendor risk, $w_4 = 0.15$ for contract risk, and $w_5 = 0.10$ for deadline urgency. These values represent the documented configuration used for the prototype and are not learned from historical compliance outcomes. Therefore, they should be interpreted as design parameters rather than statistically validated estimates of relative compliance importance.

The resulting R_{overall} value is mapped to predefined risk bands. Under the current prototype configuration, a value of A score of 0.70 or above is classed as High risk, and anything below that falls into the Medium or Low band. categories according to the configured thresholds.

The weighted contribution of module i is calculated as

$$C_i = w_i S_i \quad (2)$$

The contributions are ranked in order to identify those with the most impact to the overall assessment. BCIE retains the underlying explanation generated by the corresponding module. Rather than just outputting a standalone risk score, the platform explains the exact reasoning behind that number. For example: 'Vendor ABC showed gst errors in 6 of their last 50 invoices' This allows the user to trace the aggregate assessment back to the operational evidence that generated the individual signal.

The aggregation layer is intentionally deterministic; BCIE does not apply an additional learned model after the spe-

cialized modules have calculated their outputs. In this way, the final risk calculation can be reproduced directly from the module signals and configured weights while retaining the source of each contribution visible to the reviewer. The approach then emphasizes traceability and inspectability of the aggregation process. The effect of the configured weights and the limitations of their validation are further discussed in Section X.

The application of this aggregation model to a concrete vendor scenario is described in Section V.

V. ILLUSTRATIVE CALCULATIONS

This section demonstrates the operation of the BCIE aggregation model and the deterministic gst validation logic using representative numerical examples.

A. BCIE Weighted Risk Score

For the Vendor ABC scenario, the normalized signals are $S_{\text{legal}} = 0.55$, $S_{\text{gst}} = 0.82$, $S_{\text{vendor}} = 0.78$, $S_{\text{contract}} = 0.70$, and $S_{\text{deadline}} = 0.65$. Using the weights defined in Section IV,

$$R_{\text{overall}} = (0.20)(0.55) + (0.30)(0.82) + (0.25)(0.78) + (0.15)(0.70) + (0.10)(0.65) = 0.721 \quad (3)$$

The resulting risk value of 0.721 falls within the current High-risk band. The corresponding Business Compliance Score

is $100(1 - 0.721) = 27.9$. The weighted contributions are retained separately, allowing the final assessment to be traced

cialized modules have calculated their outputs. In this way,

to the individual module signals.

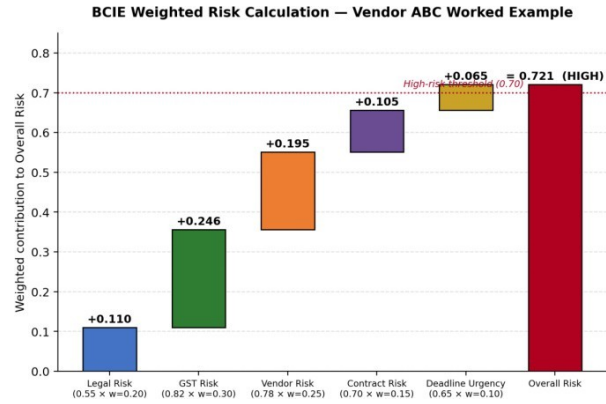
Fig. 5. Weighted contributions to the Vendor ABC overall risk score.

B. gst Rate Validation

The gst module relies on fixed arithmetic rules to check tax rates and amounts. For an invoice with a taxable value of 80,000 and an applicable 18% rate, the expected gst is

$$gst = 80,000 \times 0.18 = 14,400. \quad (4)$$

For an intra-state supply, this corresponds to $Cgst = 7,200$ and $Sgst = 7,200$. If the state 28% rate is applied, the



calculated gst becomes 22,400, producing an 8,000 difference. The module flags this condition as a rate mismatch.

These examples demonstrate the two principal functions of the framework: BCIE combines heterogeneous risk signals into a traceable business-level assessment, while deterministic gst rules provide directly verifiable statutory calculations.

VI. EXPERIMENTAL METHODOLOGY

We evaluated the prototype at two levels: how well each module performs on its own, and how well the modules work together once they are connected through the BCIE. Module-level evaluation ensures that each module delivers the expected output to accomplish its task. End-to-end evaluation ensures that the BCIE correctly synthesizes, normalizes, and explains the multiple outputs coming from the different modules. This separation allows the contribution of the proposed architecture to be evaluated independently from the performance of the individual algorithms used within the modules.

A. Experimental Setup

The evaluation was conducted using the prototype implementation described in Section III. The software environment consists of Python-based backend services, FastAPI, PostgreSQL, FAISS, OCR and natural-language processing components, and machine-learning libraries used by the individual modules. The experiment uses constructed compliance data because access to a sufficiently large anonymized enterprise compliance corpus was not available. The test data were therefore designed to reproduce representative compliance conditions while maintaining known expected outcomes for evaluation.

B. Test Data and Evaluation Scope

The legal-document testing procedure makes use of publicly available template agreements, as well as synthetically created documents with controlled modifications, such as removal of mandatory clauses, introduction of contradictions between related provisions, and near-duplicates. The supported document classes include employment, non-disclosure, vendor, and service agreements. gst evaluation makes use of constructed invoice records which contain both valid and deliberately corrupted cases. The test cases included gst rate mismatches, incorrect Cgst, Sgst, and Igst splits, duplicate and near-duplicate invoices, and invoices with missing details. The gst liability prediction experiment makes use of monthly historical records which contain sales, purchases, output gst, input gst, Input Tax Credit, and tax-paid values. We introduced a synthetic temporal variability to the data, thus obtaining temporal patterns of repeated growth and seasonality. For the vendor, contract, and deadline evaluations, we used handwritten records, which covered a variety of both risk levels and urgency constraints.

C. Compliance Test Scenarios

The evaluation is organized into eleven structured scenarios covering the principal functions of the prototype. The scenarios include classification of legal documents,

missing clauses and contradictions between the contracts, document similarity check, calculation of gst and validation of tax-split, finding exact and near duplicate invoices, identification of ITC impact, gst liability forecasting, vendor risk assessment and urgency check for contracts and deadlines. One end-to-end BCIE scenario is included, which contains various risk conditions. For the end-to-end scenario, the system receives a combination of gst, vendor, and contract signals corresponding to a high-risk business condition. The BCIE score, risk band, weighted contributions, and stated reasons are then checked against the results expected from the configured model.

D. Evaluation Metrics

Classification tasks, including document-type classification and vendor-risk classification, are evaluated using accuracy, precision, recall, and F1-score. Detection tasks, including missing-clause identification, contradiction detection, document similarity, and duplicate-invoice detection, are evaluated using precision and recall against the known set of planted or predefined test conditions. gst calculation and tax-split validation are evaluated as deterministic correctness checks. A test case is considered correct when the computed tax values and applicable Cgst/Sgst/Igst allocation agree with the expected rule-based result. We evaluated duplicate-invoice screening on cases where the exact and near-duplicate invoices were already known. We evaluated gst liability prediction with three error measures: mean absolute error (MAE), root mean squared error (RMSE), and mean absolute percentage error (MAPE). Linear Regression, Random Forest, and XGBoost are evaluated under the same time-series validation procedure, allowing their prediction errors to be compared under a common experimental setting. BCIE is evaluated separately from the module-level prediction metrics. The aggregation calculation is verified by recomputing the weighted score from the recorded module signals. Explanation consistency is assessed by checking whether the highest-ranked reported reasons correspond to the module contributions that actually determine the aggregate score.

E. Experimental Procedure

Each test case is processed through the corresponding module using the same configured rules and model versions throughout the evaluation. The predicted or detected output is compared with the predefined expected outcome, and the resulting metric is recorded. For BCIE evaluation, the module outputs are stored before aggregation so that the final score can be independently recomputed and its individual contributions traced. No modification of the expected labels is performed based on the obtained results. The purpose of the evaluation is to determine whether the implemented system behaves consistently with its predefined rules, test conditions, and expected outcomes. The resulting measurements are reported in Section VII.

VII. RESULTS AND EVALUATION

The prototype was evaluated on classification, detection, gst prediction, and BCIE aggregation tasks. The document

classification module reached 100% accuracy: missing-clause detection had a recall of 95.4%. Duplicate-invoice detection was also found to have 100% accuracy while the vendor-risk scoring was the weakest result with 57.3% accuracy.

Task	Accuracy	Precision	Recall	F1
Document classification	100.0%	100.0%	100.0%	100.0%
Missing-clause detection	—	28.3%	95.4%	43.7%
Contradiction detection	77.5%	100.0%	35.7%	52.6%
Document similarity	81.0%	64.2%	100.0%	78.2%
Duplicate invoice detection	100.0%	100.0%	100.0%	100.0%
Vendor risk scoring	57.3%	38.2%	41.3%	39.7%

Fig. 6. Classification and detection results.

For gst liability prediction, Linear Regression achieved the lowest reported error with an MAE of 4,356 and MAPE of 4.74%. BCIE aggregation was verified using the Vendor ABC scenario, producing $R_{\text{overall}} = 0.721$ and a Business Compliance Score of 27.9.

Model	MAE (₹)	RMSE (₹)	MAPE
Linear Regression	4,356	5,402	4.74%
Random Forest	5,504	6,616	6.15%
XGBoost	6,613	7,806	7.39%

Fig. 7. gst liability prediction results.

The results show that performance varies across compliance tasks. Deterministic duplicate and document-classification checks achieved high scores, while clause, contradiction, and vendor-risk tasks showed greater variation. This indicates that the reliability of the overall compliance assessment depends on both the quality of individual module outputs and the transparent aggregation performed by BCIE.

VIII. DISCUSSION

Our testing proves that rule-based checks and machine learning models can run smoothly within one architecture. We rely on deterministic logic for gst calculations, contract expiries, and statutory deadlines since those tasks have clear, absolute answers. Conversely, we reserve machine learning for areas needing pattern recognition, like document classification, semantic matching, and predicting gst liability. The BCIE brings these outputs together using a straightforward weighted-sum formula, letting us easily track each module's impact on the final score. That said, this simple linear approach is a limitation because static weights cannot capture the complex, over-lapping relationships between various risk factors. Upgrading to learned aggregation could solve this, but it adds significant complexity. That route really only makes sense once you have enough real-world training data from an actual business environment. As of now, the architecture covers legal, gst, vendor, contract, and deadline risks, while normalized scores of each module contribute to the business-level assessment. Therefore, the approach could be expanded to include other areas within a company, such as payroll liabilities, licensing, and workplace safety. The results also indicate that a unified compliance architecture should not require every module to use the same analytical technique. Deterministic validation is

appropriate where expected outcomes are explicitly defined, while learned methods are useful for classification, semantic comparison, and forecasting. BCIE provides the common interface between these heterogeneous components.

IX. DESIGN IMPLICATIONS AND CONSIDERATIONS

A. Deterministic and Learned Components

LexTax AI separates deterministic compliance operations from learned analytical tasks. We rely on hardcoded logic to manage standard checks like gst calculations, tax splits, contract expiries, and deadlines. Meanwhile, the more complex features such as document classification, semantic matching, and predicting gst liability depend entirely on machine learning and embedding models. This separation keeps rule based results directly verifiable.

B. Explainability

BCIE retains individual module signals and their weighted contributions, allowing the overall risk assessment to be traced to its underlying factors. This provides reviewers with the specific causes associated with a reported risk rather than only a final numerical value.

C. Module Independence

Each module communicates with BCIE through a standardized normalized signal and supporting information. This allows individual models or processing components to be modified or replaced without changing the core aggregation mechanism and supports the addition of future compliance modules.

D. Data and Regulatory Dependency

Vendor-risk scoring and gst liability prediction require substantial quantities of historical data to be reasonably informative, while gst validation depends on current tax rates. For these reasons, when there is insufficient historical data, such signals are not very useful. On the other hand, any changes in gst rules necessitate a continuous update of the underlying rules in the validation process.

E. Cross-Module Integration Analysis

The module-level results show that the errors differ from one compliance task to another. High results in deterministic validation and duplicate detection indicate that rule-based checks can be evaluated directly against predefined conditions, whereas clause, contradiction, and vendor-risk tasks remain more dependent on the characteristics of the underlying data. These differences are a good reason to separate each specialized analysis from the BCIE's aggregation layer. At the aggregate level, the BCIE takes the various forms of output from each module and turns it into a single representation. In the Vendor ABC scenario, the combined signals produce $R_{\text{overall}} = 0.721$, with gst and vendor risk contributing 0.246 and 0.195, respectively. The contribution values can be recomputed directly from the configured weights and module signals, allowing the final assessment to be traced to its sources. This

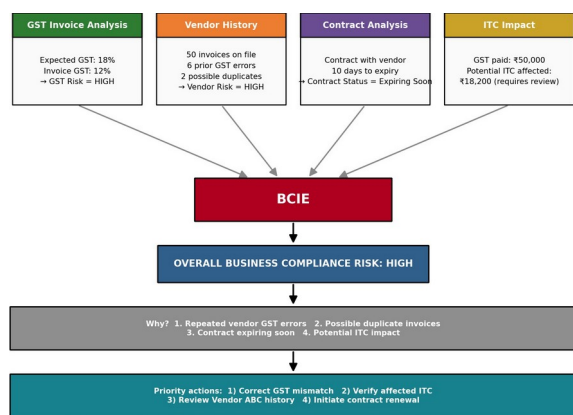


Fig. 8. BCIE end-to-end Vendor ABC risk assessment and explanation.

demonstrates the intended relationship between module-level analysis and business-level compliance assessment without treating the aggregate value as a learned probability.

X. LIMITATIONS

The reported results represent a prototype evaluated using synthetic and template-derived data rather than real business records at production scale. Legal evaluation covered a limited set of document types and constructed test cases, so performance on other document types, jurisdictions, and drafting styles remains untested. gst validation used fabricated invoices, while gst liability prediction used a synthetic monthly time series, which may not fully represent the variability of real business data. Vendor-risk scoring is rule-based and has not been validated against actual audit outcomes; therefore, it should be treated as a risk-indication mechanism rather than a predictor of future compliance failures. Similarly, legal clause and contradiction detection is intended to support document review, not provide legal advice. Drafts produced by the system, along with any issues it flags, should be checked by a qualified professional before anyone relies on them in practice.

XI. CONCLUSION AND FUTURE WORK

This paper described the LexTax AI prototype Business Compliance Intelligence Platform utilizing the BCIE as a transparent aggregation layer, which accumulated signals from the legal, gst, vendor, contract, and deadline modules, and distilled them down to the business-level compliance assessment. The value of this prototype was largely determined by its architectural novelty, which underpinned the combined application of the tools used. The signals processed by the system underwent normalization and were aggregated into a final business-level compliance assessment using a deterministic weighted model, the parameters of which remained available for inspection.

The value of such an approach was primarily determined by the fact that the business level compliance assessment used

different signals that originated from the separate analyses performed by the system. In the case of deterministic checks, such as gst and deadlines, the processing of these signals yielded results that could be interpreted as standalone compliance assessments. At the same time, analyzing the legal, vendor, and gst forecasts gave us a much deeper context for evaluating the business as a whole. We specifically designed the architecture to separate raw compliance detection from actual decision making. Each module has one core job: flagging specific issues like tax mismatches, absent clauses, duplicate files, or tight deadlines. From there, the BCIE steps in to consolidate everything. Keeping these steps distinct means we preserve the evidence for every single flag, rather than just spitting out a generic, unexplainable final score. Because of this, our project serves purely as an advisory tool. It is built to speed up human reviews, not replace human judgment or make autonomous choices.

Looking ahead, we have three primary goals for future development. First, we want to test learned aggregation methods on real audit data to see how they hold up in practice. Second, we plan to refine our vendor risk models as we get our hands on more labeled historical records. Finally, we need to implement a versioned gst database to ensure our rules always match the latest tax regulations. Since this is currently a prototype, our immediate next step involves rigorous testing against larger, anonymized corporate datasets alongside extensive user feedback to truly validate its real world effectiveness.

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